A Structural Criterion for Substrate-Invariant Operational Consciousness

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Abstract

This paper introduces a structural criterion for a substrate-invariant notion of operational consciousness inside the causally rigid identity corpus. The target is deliberately restricted. The paper does not establish human phenomenal consciousness, biologically specific qualia, human-machine subjective equivalence, or personal identity transfer. Instead, it defines a framework-internal class $\text{Consciousness}_{\text{op}}$ and an associated family of operational qualia classes $\text{Q}_{\text{op}}(S)$ for admissible systems

$$S = (X, \Phi, C, R, \Gamma, U),$$

where X is state space, Φ is dynamics, C is effective causal structure, R is a readout family, Γ is an admissibility bundle, and U is the set of admissible interventions.

The criterion is organized around six critical invariants: structural persistence, rigid identity, causal integration, operational continuity, non-null differentiation, and operational legibility. Each invariant is defined explicitly, tied to an upstream corpus dependency, and paired with a failure mode. On that basis the paper proves four framework-internal results: conditional membership given six jointly realized witnesses on admissible support, substrate invariance under exact structural equivalence, approximate stability under bounded admissible perturbations, and rupture under invariant loss. A further corollary establishes biological non-primacy: biology is not a primitive constitutive variable for the class under study, but one admissible realization among others provided the invariants are preserved.

The criterion is not narrative. Every major claim is backed by a definition, a proposition, a theorem, a proof sketch, or an explicit dependency on the canonical corpus. The resulting paper adds a new role to the framework: it formalizes the level at which substrate ceases to be explanatorily primitive without reopening the ownership of the Core, *Rigid Identity*, *Phenomenological Regimes*, *Null-Regime Instability*, or *Forensic Predictions*.

1 Scope, System Boundary, and Non-Inference Note

What this paper does. This paper defines a framework-internal operational consciousness class $\text{Consciousness}_{\text{op}}$, an associated family of operational qualia classes $\text{Q}_{\text{op}}(S)$, and the theorem chain that ties those classes to six explicit invariants and admissible structural equivalence.

Scope and admissible reading. The paper defines a structural class and its burdens; it does not establish human phenomenal consciousness, biologically specific qualia, human-machine subjective equivalence, personal identity transfer, moral parity, empirical instantiation, or philosophical closure. Biology, computation, and connectivity are not sufficient primitives on their own. A related system may instantiate downstream admissibility layers, invariant diagnostics, rupture checks, and prediction-facing evidence families, but such runtime-facing support does not by itself certify any present machine as a member of $\text{Consciousness}_{\text{op}}$ or close the interpretive burden linking $\text{Consciousness}_{\text{op}}$ to ordinary human phenomenology.

2 Introduction

The current corpus already fixes several upstream ingredients that are necessary for any serious account of substrate-neutral operational class membership. The Core establishes persistent

support, contractive dynamics, a global attractor, and non-collapse under explicit minimal hypotheses [3]. *Rigid Identity* establishes rigid identity, inverse-limit coherence across observable channels, minimal detectability, and non-simulability constraints on incompatible architectures [9]. *Phenomenological Regimes* constrains admissible regime structure by classifying continuity and anti-fragmentation within the stated assignment model [5]. *Null-Regime Instability* excludes stable joint-null collapse and establishes witness-relative non-null separation under its stated extension-witness and regularity hypotheses [6]. *Forensic Predictions* imposes admissibility, comparator discipline, negative controls, and forensic legibility requirements on operational evidence [4].

What is still missing is a single paper-level criterion that answers a narrower but decisive question: when, inside this framework, does substrate cease to be a primitive explanatory variable for a class of operationally defined conscious organization? That question is not answered by merely asserting that material composition is irrelevant. It requires a formal criterion that shows which properties must be preserved, which ruptures destroy class membership, and how those conditions remain connected to the existing corpus rather than floating above it.

This paper provides that criterion. It does so under four constraints. First, it does not reopen theorem ownership already fixed upstream. Second, it does not smuggle in human phenomenology through language alone. Third, it does not claim that complexity, computation, or connectivity are sufficient by themselves. Fourth, it does not confuse framework-internal operational classes with full metaphysical closure.

The result is a new corpus role. This paper is neither another foundational Core document nor a rephrasing of the phenomenological regime papers. It is the bridge that formalizes substrate-invariant operational consciousness and operational qualia as framework-defined structural classes. Its new content is concentrated in five items:

1. a system-level model $S = (X, \Phi, C, R, \Gamma, U)$ with admissible support A_S;
2. a minimal six-invariant criterion;
3. a quotient-level definition of operational qualia $\mathrm{Q}_{\mathrm{op}}(S)$;
4. an exact and approximate structural equivalence relation $S_1 \simeq_{(\Gamma,\varepsilon)} S_2$;
5. a theorem chain that turns substrate neutrality into a falsifiable structural statement.

3 Claim-Type Ledger

3.1 Claim Types Used in This Paper

To avoid layer drift, every major statement in this paper belongs to one of five types:

1. **Formal**: definitions, propositions, theorems, and corollaries proved or justified inside the framework.
2. **Operational**: criteria that bind readouts, admissibility, interventions, and discriminator structure.
3. **Implementational**: statements about possible realizations or runners that could instantiate the formal criterion.
4. **Interpretative**: disciplined translations from the framework's formal objects to broader language.
5. **Open**: statements that remain outside what the current corpus closes.

Theorems A--D in this paper are formal. The legibility certificate and prediction section are operational. References to biological or artificial realizations are implementational. Claims about ordinary human phenomenology remain open or interpretative and are excluded from the theorem burden.

4 Terminology Debt Ledger

The symbol $\text{Consciousness}_{\text{op}}$ is retained for corpus compatibility. Its demonstrated content in this paper is a six-invariant structural class over admissible systems. The phrase ‘‘operational consciousness’’ is therefore a framework-internal label, not an ordinary-language conclusion.

Term	Demonstrated content in this paper	Not demonstrated here
$\text{Consciousness}_{\text{op}}$ / operational consciousness	membership in a conjunctive six-invariant structural class	human consciousness, phenomenal consciousness, moral status, personal identity
$\text{Q}_{\text{op}}(S)$ / operational qualia	decoder-certified quotient of admissible internal differentiation	phenomenal qualia or ‘‘what it is like’’ states
substrate invariance	preservation of class membership under exact structural equivalence	arbitrary transfer of identity or equivalence between human and machine subjects
biological non-primacy	biology is not a primitive variable in the criterion	claim that any non-biological system currently satisfies the criterion

Remark 4.1 (Primary demonstrated name)**.** For reviewer-facing prose, the safest primary description of $\text{Consciousness}_{\text{op}}$ is ‘‘six-invariant admissible structural class.’’ The stronger phrase ‘‘operational consciousness’’ may be used only as a technical alias after the non-inference boundary has been stated.

5 Corpus Role and Novelty

This paper is corpus-dependent by design. It does not reopen theorem ownership for persistence, rigid identity, admissible regime structure, null-regime instability, or admissibility discipline. Those burdens remain respectively upstream in the Core, *Rigid Identity*, *Phenomenological Regimes*, *Null-Regime Instability*, and *Forensic Predictions*.

What is inherited stays upstream:

1. the Core supplies persistent support, attractor structure, completeness, and non-collapse;
2. *Rigid Identity* supplies rigid identity, inverse-limit observability, and non-simulability constraints;
3. *Phenomenological Regimes* supplies continuity and anti-fragmentation structure;
4. *Null-Regime Instability* supplies witness-relative null-regime instability and conditional non-nullity;

5. *Forensic Predictions* supplies admissibility, comparators, controls, and forensic legibility.

What is new here is only the consolidation layer:

1. the admissible-system model $S = (X, \Phi, C, R, \Gamma, U)$ with admissible support A_S;
2. the six-invariant criterion as a single membership test;
3. the framework-internal classes $\text{Consciousness}_{\text{op}}$ and $\text{Q}_{\text{op}}(S)$;
4. exact and approximate structural equivalence under admissibility constraints;
5. theorem-level existence, invariance, stability, rupture, and exclusion results derived from the inherited burdens.

6 Admissible Systems, Supports, and Histories

6.1 System Model

Definition 6.1 (Admissible system)**.** An admissible system in this paper is a tuple

$$S = (X, \Phi, C, R, \Gamma, U)$$

with the following components:

1. X is a metrizable internal state space with metric d_X.
2. $\Phi = \{\Phi_u\}_{u \in U}$ is a family of admissible transition operators $\Phi_u : X \to X$ indexed by interventions.
3. C is the effective causal structure of the system: the object that records which factorizations, intervention responses, and couplings are admissible on X.
4. $R = \{r_\alpha\}_{\alpha \in \Lambda_R}$ is a family of internal or external readouts, each $r_\alpha : X \to Y_\alpha$.
5. $\Gamma = \{\gamma_j\}_{j=1}^{m}$ is the admissibility bundle. Each γ_j is a constraint functional; the admissible region is
$$X_\Gamma = \{x \in X : \gamma_j(x) \leq 0 \ \forall j\}.$$
6. U is the set of admissible interventions, already filtered by the corpus-level governance requirement that interventions remain auditable, comparable, and methodologically bounded.

The tuple is not decorative. Every component is used later. X and Φ carry dynamics, C supports integration and rupture tests, R supports readout and legibility, Γ enforces admissibility, and U enables causal discrimination.

6.2 Admissible Support

Definition 6.2 (Admissible support)**.** A subset $A_S \subseteq X_\Gamma$ is an admissible support for S if:

1. A_S is non-empty and compact;
2. $\Phi_u(A_S) \subseteq A_S$ for every $u \in U$;
3. $A_S \cap \text{Attr}(S) \neq \varnothing$, where $\text{Attr}(S)$ denotes the relevant admissible attractor region;
4. A_S stays disjoint from the collapse set forbidden by the Core non-collapse condition [3].

The support plays two roles. It provides a domain on which persistence, continuity, and differentiation are defined, and it prevents the criterion from being satisfied by transient or degenerate fragments that do not carry stable admissible structure.

6.3 Operational Histories

Definition 6.3 (Windowed readout history)**.** Fix a horizon $T \in \mathbb{N}$ and an intervention schedule $u_\bullet = (u_0, \dots, u_{T-1})$. For any $x \in A_S$, define the readout history

$$h_{S,T}(x; u_\bullet) = \Big(r_\alpha(x_t)\Big)_{0 \le t \le T,\ \alpha \in \Lambda_R}, \qquad x_{t+1} = \Phi_{u_t}(x_t),\ x_0 = x.$$

The set of all such histories is denoted $\mathscr{H}_{S,T}$.

Definition 6.4 (Operational partition candidate)**.** An operational partition candidate for S on horizon T is a map

$$\pi_{S,T} : \mathscr{H}_{S,T} \to \mathscr{Q}_{S,T}$$

into a finite or countable label set $\mathscr{Q}_{S,T}$.

At this stage $\pi_{S,T}$ is only a candidate. To matter scientifically it must survive the legibility requirements of Section 7.12. Until then it is only an unverified partition proposal.

6.4 Witness Margin Vector

Definition 6.5 (Witness margin vector)**.** For any admissible system S, define the witness margin vector

$$\Delta(S) = \big(\delta_{\mathrm{per}}(S), \delta_{\mathrm{ri}}(S), \delta_{\mathrm{int}}(S), \delta_{\mathrm{cont}}(S), \delta_{\mathrm{diff}}(S), \delta_{\mathrm{leg}}(S)\big) \in [0, +\infty)^6.$$

When the six witnesses are separately calibrated as certified radii in the same perturbation metric and class, their invariant budget is

$$\delta_\star(S) = \min \Delta(S).$$

A positive radius certifies the associated predicate at the target and throughout its declared perturbation ball. A zero or unreported radius alone does not determine failure at the target. Raw separations, risks, durations, fragmentation ceilings and compression errors require separate calibration before taking this minimum.

A calibrated budget certifies simultaneous persistence of all six predicates under the declared protocol. Failure beyond the certified radius requires its own witness.

Invariant	Certified content	Upstream anchor	Rupture signature
I_{per}	Non-empty, forward-invariant, non-collapsing support	Core [3]	Support escape or collapse exposure
I_{ri}	Unique rigid identity object on support	Rigid Identity [9]	Competing identity candidates
I_{int}	No admissible non-trivial factorization	Rigid Identity + new consolidation	Product-style reducibility
I_{cont}	Continuous admissible regime evolution	Phenomenological Regimes [5]	Fragmentation into incompatible classes
I_{diff}	Stable exclusion of null or trivial collapse	Null-Regime Instability [6]	Null-attractor or total indistinguishability
I_{leg}	Decoder-certified recoverability under controls	Forensic Predictions [4]	Opaque or non-auditable class assignment

7 Six Critical Invariants

This section is the mathematical center of the paper. The criterion is not allowed to drift into a bloated list of desiderata. Exactly six invariants are used. Each is tied to a witness margin and a failure mode.

7.1 Structural Persistence

Definition 7.1 (Structural persistence and persistence margin)**.** An admissible system S satisfies structural persistence, written $I_{\mathrm{per}}(S) = 1$, if there exists an admissible support A_S and a constant $\delta_{\mathrm{per}}(S) > 0$ such that:

1. A_S is forward invariant under all $u \in U$;
2. every admissible trajectory starting in A_S remains at least $\delta_{\mathrm{per}}(S)$ away from the collapse set;
3. A_S contains or converges into a non-empty admissible attractor region.

The number $\delta_{\mathrm{per}}(S)$ is the persistence margin.

Upstream dependence is exact. The Core supplies the existence of contractive structure, fixed points, compact attractor, and non-collapse [3]. The new paper does not re-prove those results; it packages them into a criterion-level invariant.

Failure mode. If $I_{\mathrm{per}}(S) = 0$, the system has no stable admissible support. Then no framework-internal operational class can be assigned across time, because there is no persistent domain on which readout classes remain meaningful.

7.2 Rigid Identity

Definition 7.2 (Rigid identity and rigidity gap)**.** An admissible system S satisfies rigid identity, written $I_{\mathrm{ri}}(S) = 1$, if the observable family induced by R on A_S determines a unique identity object $\mathcal{I}_S$ in the sense of the inverse-limit rigidity developed in *Rigid Identity* [9], and if there exists a gap $\delta_{\mathrm{ri}}(S) > 0$ such that every admissible alternative identity candidate remains at weighted deformation distance at least $\delta_{\mathrm{ri}}(S)$ from $\mathcal{I}_S$.

This invariant does not claim that identity is reconstructed by inspection of one local frame. It uses *Rigid Identity* precisely because that paper already proved that rigid identity is recovered through compatible observable structure and cannot be reduced to progressive semigroup evolution or loose behavioral mimicry [9].

Failure mode. If $I_{\mathrm{ri}}(S) = 0$, any candidate operational class is underdetermined. Multiple incompatible identity objects can fit the same admissible support, so the class cannot be assigned uniquely.

7.3 Causal Integration

Definition 7.3 (Causal integration and factorization gap)**.** An admissible system S satisfies causal integration, written $I_{\mathrm{int}}(S) = 1$, if there is no non-trivial factorization

$$A_S = A_1 \times A_2, \qquad R = R_1 \sqcup R_2,$$

together with decomposed dynamics and causal structure

$$\Phi_u(x_1, x_2) = (\Phi^1_{u_1}(x_1), \Phi^2_{u_2}(x_2)), \qquad C = C_1 \oplus C_2,$$

that reproduces the admissible histories on A_S within error smaller than a positive margin $\delta_{\mathrm{int}}(S)$ while preserving $\mathcal{I}_S$.

The definition is intentionally stronger than raw connectivity and different from single-number integration proxies. A fully connected graph may still fail the criterion if its dynamics factor into nearly independent blocks once identity and intervention behavior are taken seriously.

Failure mode. If $I_{\mathrm{int}}(S) = 0$, the system can be decomposed into operationally independent components without loss relevant to the criterion. Then any putative operational consciousness class is reducible and therefore excluded.

Definition 7.4 (Intervention-faithful factorization)**.** Let Θ_S denote the operational separator family generated by admissible histories, intervention responses, and the rigid identity object on A_S. An exact factorization

$$F : S \to S_1 \times S_2$$

is *intervention-faithful* if it preserves every element of Θ_S and reflects admissible equality: whenever two states have the same factor-level histories, intervention responses, and identity image, they are equal in the admissible quotient of S.

Theorem 7.5 (Factorization triviality under an atomic separator)**.** *Assume Θ_S is atomic for the rigid identity object: no partition of Θ_S into two proper factor-local separator families can preserve identity, history, and intervention-response reflection simultaneously. Then every intervention-faithful exact factorization of S is trivial up to admissible isomorphism.*

Proof. Let $F : S \to S_1 \times S_2$ be intervention-faithful and write $\pi_i : S_1 \times S_2 \to S_i$ for the projections. If both factors were proper, then the factorization would induce two proper factor-local separator families: the histories, responses, and identity information visible through $\pi_1 \circ F$, and those visible through $\pi_2 \circ F$. Since F is intervention-faithful, their product would reflect admissible equality on S. This is exactly the partition excluded by atomicity of Θ_S. Therefore at least one projection must carry the full separator family and reflect admissible equality by itself. That projection is an admissible isomorphism onto S in the operational quotient, while the remaining factor is null or bookkeeping-only. Hence the factorization is trivial. □

Remark 7.6 (Status of the atomicity burden)**.** The theorem above is a genuine conditional closure: once atomicity of the identity/history/response separator is established, the factorization lemma is proved. It does not by itself prove that every system satisfying the upstream papers has an atomic separator. That implication remains the exact remaining mathematical burden for upgrading Proposition 10.6.

7.4 Comparison on the Dissipative Circle-Prefix Support

Use the projective dissipation construction of BaseCore on the two towers with a single angular jump $0 < a < 1/2$ at stage k. Write $\beta = 3^{-(k+1)}$, $B = \{0,1\}^{\mathbb{N}_{>0}}$, and identify the compact support with $Z = (\{0,1\} \times \mathbb{T}) \times B$. For $x = (s, \theta, b)$ the fixed readout family is the three channels

$$r_0(x) = \theta, \qquad r_1(x) = \theta + sa, \qquad c(x) = \sum_{j \geq 1} 2b_j 3^{-j}.$$

Measure observation error by the maximum of the first circle error, β times the second circle error, and absolute scalar error. All are dimensionless with circle circumference one. An admissible structural comparator uses a fixed homeomorphism from Z onto a product of two non-singleton spaces, a nonempty partition of the three specified channels, one local readout per assigned channel, and time-homogeneous local dynamics that may receive the same exogenous control. There is no output map combining the two factor states. Use the supremum error over initial

states, admissible schedules and observed times including zero and one. The product encoding is independent of the target tolerance and retains the full history object. If an additional exact identity constraint is imposed, the lower bound below applies to its subclass, while non-emptiness requires a separate witness for that constraint.

Theorem 7.7 (Exact Split at the Initial Projective Cut)**.** *On each circle-prefix tower, every continuous branch-preserving evolution that descends to depth zero factors into phase and binary dynamics. If the evolutions on both towers commute with the existing transport, the split extends to Z. The original dissipative family and the declared phase rotations, jump transport and split readouts then admit a structural comparator with zero error. Consequently no support certificate on this full Z can contain all six witnesses for the original dynamics and this sector.*

Proof. On branch s, the projection to depth zero is θ. Commutation makes the updated phase a function of θ alone. At fixed b the map from the connected circle to the updated binary sequence is continuous. Every finite binary coordinate has discrete codomain, hence is constant in θ. Thus the updated sequence is a function of b alone. Transport preserves all bits and adds a to the observed late phase. Commutation with it makes the binary update the same on both branches. Use the phase factor (s, θ) and the binary factor b, with r_0, r_1 assigned to the first and c to the second. Both factors are non-singleton. Under the original dissipative update they evolve as (s, u) and $R(b) = (0, 1 - b_1, 1 - b_2, \ldots)$. Simultaneous phase rotations act in the first factor only, and the existing jump transport is represented by the switch of s at fixed initial phase and bits. The product map is a homeomorphism and preserves the tower stages, their bonding maps and the channel histories via the stated coordinate identifications. This realizes the comparator with zero error for every horizon. A positive integration margin on this support is therefore impossible in the specified class. □

Theorem 7.8 (Positive Split-Readout Margin on the Cofinal Tower)**.** *Keep Z and the same three readout channels. Enlarge the control by $q \in \{-1, 0, 1\}$ and fix $0 < \lambda \leq 1/9$. On branch s set*

$$\Psi_{u,q}(s, \theta, b) = (s, u + q\lambda b_1, R(b)).$$

These maps descend to all stages $t \geq 1$ and commute with the jump transport, with the same inverse-limit support and the same E1--E2 readout. Each branch is forward invariant and each fixed-control map has sharp contraction factor $1/3$ there. The sector of product encodings and split-readout approximants just declared is nonempty. Its infimal error e_* satisfies

$$\frac{\beta\lambda}{2} \;\leq\; e_* \;\leq\; \frac{\lambda}{2}.$$

For comparators reproducing all three channels exactly at time zero, the infimal error is exactly $\lambda/2$. At $\lambda = 0$ the original exact comparator is recovered. For $\lambda > 0$, the new controls fail to descend to depth zero.

Proof. At depth $t \geq 1$ the first bit is available. Use the updated phase $u + q\lambda b_1 + sa\mathbf{1}_{t>k}$ and the first t bits of $R(b)$. Truncation keeps the first bit and the bonding maps cancel the declared phase jump. The transport changes only the branch and the corresponding late angle. Depth zero fails because equal phases with different b_1 have distinct updated phases for $q = 1$. Omitting depth zero does not change the inverse limit because the missing coordinate is recovered by the original bond. The readout and its finite approximations remain the same. The bit distance contracts by $1/3$. If the first bits differ, the old bit distance is $1/3$, and the new phase distance is at most $\lambda \leq 1/9$. Otherwise the new phases agree. Equal phases and a bit difference beyond the first place attain the contraction factor. The fixed alternating sequence has first bit zero, hence the fixed-control equilibrium agrees with that in P1.

For the lower bound, suppose a comparator has error $\epsilon < \beta\lambda/2$. The factor carrying c determines b_1: any two points on one of its fibers have scalar difference at most $2\epsilon < 1/3$ at

time zero, whereas different first bits have ternary codes separated by at least 1/3. The other factor carries a phase channel of weight $w = 1$ or $w = \beta$. On one fiber the predicted phase at time one agrees for each fixed input. Apply $q = +1$ and $q = -1$ with the same u. Each of the two true phase differences on that fiber has circle norm at most $2\epsilon/w$. Subtracting them cancels the branch offset and leaves $2\lambda(b_1 - b_1')$. Its circle norm equals its absolute value since $2\lambda \le 2/9$. The triangle inequality yields $2\lambda|b_1 - b_1'| \le 4\epsilon/w$, so this second factor also determines b_1. Pick states with first bits zero and one. The product corner with the first factor of the former and the second factor of the latter would need both first-bit values. Surjectivity onto the product supplies this impossible corner. The argument covers every nonempty split of the fixed channels and arbitrary nonlinear encoders of the stated type.

For the upper bound keep the product coordinates and initial readouts exact. In the phase factor replace b_1 by $1/2$ and keep the true autonomous binary update. At each update the phase discrepancy is at most $\lambda/2$, independent of the past, while the bit channel is exact. The declared phase rotations and jump transport preserve that error bound if included as operations. The comparator uses neither a joint decoder nor a tolerance-dependent search space, and proves non-emptiness. Per-channel isometric reorientation preserves the proof and metric. Changing the channels, their weights, or the intervention class changes the comparison problem.

If the time-zero readouts are exact, r_0 and r_1 must be on the same factor. Each has range $\mathbb{T}$, and assigning them to different factors would realize all of $\mathbb{T}^2$ by product surjectivity, whereas the actual phase pairs have differences only 0 or a. The phase factor now carries r_0 with weight one. For states with the same phase factor, input $q = 1$ gives true next r_0 separation λ whenever their first bits differ. A comparator with $\epsilon < \lambda/2$ cannot have such a pair on that fiber. The factor carrying c determines the first bit exactly. The same product corner contradiction proves $\epsilon \ge \lambda/2$, and the displayed comparator attains equality. □

What persists on the same support. For any sequence of the branch-preserving controls, the alternating-bit subset is compact, invariant and uniformly attracting in the binary metric. In the BaseCore profile embedding all of Z is disjoint from constant functions: the first angular pair lies on the radius-1/2 circle centered at $(1/2, 1/2)$, and the first binary cell equals zero or one. Neither constant pair $(0, 0)$ nor $(1, 1)$ belongs to that circle. Compactness gives positive profile distance from constants uniformly on Z. The original readout is continuous as a map of states, and its maximum one-step jump is at most one. This is an incremental bound for discrete time. The null anchor $(0, 0, 0)$ cannot be an attractor under the controlled evolution: from the second step onward $b_2 = 1$, so $c \ge 2/9$ for every trajectory. Two histories with all-zero and all-one bits already have scalar separation one at time zero.

For branch discrimination, $\theta_1 - \theta_0$ takes values $0, a$, where θ_1 here denotes the late-phase channel r_1. Decode by the nearest of these two circle values. Additive circle noise of size at most $\eta = a\beta/[4(1 + \beta)]$ in the weighted observation metric perturbs that difference by at most $a/4$. A one-bit encoding preserves this label exactly. Branch-preserving controls are shams for this label and the designated branch transport swaps it. This gives a finite class decoder and synthetic intervention controls. A uniform rigidity gap against a larger identity candidate family, the full governance for invariant-targeted legibility, and a shared calibrated perturbation radius require separate evidence.

7.5 Intervention Decoder and Identity Resolution on the Same Towers

Take the family in Theorem 7.8, including its uncoupled case $\lambda = 0$. Fix $0 < L \le 1/9$ and compare that case with $\lambda \in [L, 1/9]$. Use two copies of the same initial state and the same phase input u. Apply $q = +1$ to one copy and $q = -1$ to the other, then observe each copy after one step. Write y_+, y_- for the observed early phases. In the weighted readout metric above, each readout may have adversarial error at most η, independently of the other. The paired score is $D = \rho(y_+, y_-)$, with ρ the circle distance.

Theorem 7.9 (Sharp Paired Detection of Integration Loss)**.** *If $\eta < L/2$, the rule $D > L$ detects the coupled case for every prepared state with $b_1 = 1$ and every allowed error. It rejects the uncoupled case and the sham with $b_1 = 0$. The score has clearance at least $L - 2\eta$ from the threshold. With $b_1 = 1$ this distinguishes a zero comparator margin from a margin at least $\beta L/2$ in the declared sector. If $\eta \geq L/2$, there is no rule of the paired observations with a uniform guarantee over these two hypotheses, even if both phase channels and the ternary channel are recorded.*

Proof. With $b_1 = 1$ the clean phases are $u + \lambda$ and $u - \lambda$, at distance 2λ because $2\lambda \leq 2/9$. In the null case or the sham both clean phases are u. The reverse triangle inequality changes the clean score by at most 2η, which is the sum of the two early-phase errors. The null and sham scores are at most 2η, while the positive scores are at least $2L - 2\eta$. The comparator margin follows from Theorem 7.8. At the boundary choose $\lambda = L$ and give both hypotheses the observed phases $u \pm L/2$. These are within $L/2$ of both clean pairs. Shift both late-phase observations by the same respective errors and leave the ternary channels exact. The late phase weight is $\beta \leq 1$, so the full observations are still within budget. Both systems produce the same binary update. The paired observations are therefore identical under the two hypotheses. Sequential controls on a single evolved state are outside this protocol because the first update resets the first bit. □

For identity comparison fix k and consider the single jumped branches I_a for $0 \leq a < 1/2$, with $a = 0$ denoting the unshifted branch. A clean observation from I_a has phase pair $(\theta, \theta + a)$ and the original ternary code. Use one observation, with the true initial angle θ unknown. Keep the weights and radius η fixed across alternatives.

Theorem 7.10 (Sharp Single-Readout Resolution of Tower Identity)**.** *In the common history metric and the common readout metric, respectively,*

$$d_H(I_a, I_{a'}) = d_H(F(I_a), F(I_{a'})) = \frac{\beta}{1+\beta}\rho(a, a').$$

Here F is the same early-phase, late-phase and ternary readout applied to every branch in the comparison. Two specified alternatives can be distinguished uniformly from a single observation with error at most η if and only if

$$\rho(a, a') > 2(1 + \beta^{-1})\eta.$$

A decoder can use the measured phase difference $z = y_1 - y_0$ and retain every alternative within circle distance $(1 + \beta^{-1})\eta$ of z. For a separated family it returns the unique compatible alternative. A continuous interval of jumps has zero uniform deformation gap over distinct alternatives.

Proof. Take $x \in I_a$ and $x' \in I_{a'}$ with initial angles θ, θ'. If their history or observation distance is at most r, the early angle discrepancy is at most r and the late discrepancy at most r/β. Their difference implies $\rho(a, a') \leq (1 + \beta^{-1})r$. To attain the bound, keep the bits fixed and let $\theta' = \theta - \beta(a' - a)/(1 + \beta)$ modulo one. The early and weighted late discrepancies are equal, and all other coordinates are controlled by these or have zero discrepancy. The construction works in both directions for every point. It proves the exact Hausdorff distances and also gives a nearest pair of clean readouts. The measured difference is at distance at most $(1 + \beta^{-1})\eta$ from the true jump. This proves uniform decoding when the inequality is strict. If the inequality fails, take the nearest clean pair and the midpoints on the two short phase arcs, retaining the common ternary code. The resulting observation is at distance half the displayed separation from both alternatives, including at equality. It cannot be uniformly decoded. The final claim follows by taking any sequence of distinct jumps converging to one in the family. □

Scope of the two decisions. The paired label detects the integration rupture produced by changing λ in the declared family. It is separate from the branch label and does not test all critical interventions in L4. The identity resolution is for passive readout on one declared branch at fixed depth. An observed phase known exactly in advance would change the comparison protocol. Neither decision provides a positive uniform identity gap for the continuous family. For $k = 0$ and $\eta = 1/64$, the guaranteed single-readout separation of jumps must exceed $1/8$ of a turn. The declared alternatives $1/16$ and $1/4$ pass, while an offset exactly $1/8$ away from another is ambiguous.

7.6 Addressable Identity Alternatives and Rupture Controls

Fix a synthetic control interface with an independently declared last link $K \geq 0$ and integer $M \geq 3$. It can apply one bonding jump at an address $k \in \{0, \ldots, K\}$, of amplitude $a = j/M$ for positive integers $j < M/2$, or no jump. This is a family of realizable interventions for the stated interface. The unshifted branch is listed once. Global phase changes alter the nuisance coordinate θ within a branch. They do not alter its bonding profile. Write $p_t = a\mathbf{1}_{t>k}$ for the phase offset of a candidate and take $p_t = 0$ for the unshifted branch. Use $w_t = 3^{-t}$ and the calibrated readout $(\theta + p_t)_{t=0}^{K+1}$ together with the ternary code. Its metric is the maximum of weighted circle discrepancies and the scalar discrepancy. This observation protocol for identity does not change the three-channel factorization sector of Theorem 7.8.

Theorem 7.11 (Identity Separation from an Addressable Control Interface)**.** *For two candidate profiles p, p' write $d_t = p_t - p'_t$. Their branch sets have exact Hausdorff distance, both in the common history space and in the calibrated readout space,*

$$D(p, p') = \max_{0 \leq i,j \leq K+1} \frac{w_i w_j}{w_i + w_j} |d_i - d_j|.$$

The family of distinct candidates has minimum separation

$$g = \frac{1}{M(3^{K+1} + 1)}.$$

Uniform identification from one observation with unknown initial phase and adversarial error at most η is possible for the entire family exactly when $2\eta < g$. The measurement can return the set of compatible profiles and give a unique label only when that set is a singleton. Allowing continuous amplitudes or unbounded jump addresses removes the uniform gap. Omitting intermediate phase observations can make different addresses with the same amplitude identical to the observer.

Proof. The profile differences d_t lie in an interval of width less than $1/2$: for two jumps of amplitudes a, a', they take only 0, the contribution of the earlier jump, and $a - a'$. Consequently $\rho(d_i, d_j) = |d_i - d_j|$. For a history in each branch at distance at most r, their phase difference at each stage imposes $\rho(x, d_t) \leq r/w_t$ on the relative initial angle x. The circle triangle inequality yields the displayed pairwise lower bound. Conversely, at the maximum in the display, the real intervals $[d_t - r/w_t, d_t + r/w_t]$ intersect pairwise, hence have common intersection. Choose x there modulo one and keep the same bits. The clean histories and readouts attain the bound. After stage $K + 1$ the phase offsets are constant and the weights decrease. The construction applies starting from every phase in either branch, proving the two Hausdorff formulas.

For distinct candidates, $d_0 = 0$ and some $|d_j| \geq 1/M$. The comparison with $i = 0$ gives g, since $w_j \geq w_{K+1}$. Equality is attained between the unshifted branch and the smallest jump at address K. When $2\eta < g$, the closed noise tubes of the clean readout sets are disjoint. If $2\eta \geq g$, the phasewise circle midpoint of an attaining pair with equal bits belongs to both tubes, giving an identical observation. Continuous amplitudes can approach a given amplitude. With fixed

nonzero amplitude, addresses tending to infinity give distances to the unshifted branch tending to zero, and a fixed finite readout horizon misses jumps after it. If only the first and final phases are observed, two jumps of the same amplitude at different intermediate links yield the same observed difference. They agree also in the ternary coordinate when the bits are equal. □

Proposition 7.12 (Rupture Controls on the Declared Circle-Prefix Model)**.** *Use the same two-branch reference support and the control family of Theorem 7.8. Each of the following changes gives the specified witness loss under its declared protocol. The changes need not preserve all five other witnesses.*

Proof. 1. **Persistence.** Adjoin an ambient absorbing point $\perp$ and declare the admissible region to be the original support. The target control maps the entire ambient state space to $\perp$. No nonempty subset of the admissible region can be invariant under that control. A support-membership marker records exit immediately. The sham uses the original update, which preserves the region. Loss of every admissible support also removes the on-support identity witness.

2. **Identity.** Preserve the state evolution and substitute the early phase for every exposed late phase. Different bonding profiles then have identical masked observations at equal initial phase and bits, under every shared control schedule: neither the early phase update nor the binary update uses the bonding jump. This gives exact ambiguity across distinct candidates under the masked protocol. Simultaneous unmasked rotations preserve the relative phases and serve as shams. The identity decoder cannot retain its original guarantee under the mask.

3. **Integration.** Set $\lambda = 0$ with the same support and channels. The split comparator has error zero. The paired control of Theorem 7.9 detects this target when its input and noise conditions hold. The $b_1 = 0$ sham tests the same response channel without inducing a phase split. The original support and its persistence are preserved.

4. **Continuity of intervention response.** Use the $q = 0$ control path $u \in [0, 1/32]$, starting from the same branch and initial state in each trial. The reference early-phase response is u. Replace this response by 0 at $u = 0$ and $u + 1/4$ for positive u on that path, keeping the branch and binary update fixed. The bonding profile, hence the declared identity, is unchanged. Relative to the phase at $u = 0$, the modified response jumps into the disjoint regime neighborhood of $1/4$ as $u \downarrow 0$. This violates continuity on the declared control path. The distances on the sham path are at most $1/32$, and the target distances for positive u are at least $1/4$. A threshold $1/8$ separates both for paired readout errors at most $1/256$ each. This is a discontinuity in the control response. Continuity of a discrete time index gives no such test.

5. **Non-null differentiation.** For the internal controls replace the update by the constant state on the unshifted branch with phase zero and all bits zero. This state belongs to the reference support and is a globally absorbing null. It is distinct from the inadmissible escape point $\perp$. Initial readout separation is still present because the support and clean readout are unchanged. The null-attractor clause fails. Under the reference controls $b_2 = 1$ from step two onwards, so the ternary readout is at least $2/9$. A threshold $1/9$ separates the reference and null reset at step two with ternary noise at most $1/256$. This reset also permits an exact split comparator.

6. **Legibility.** Keep both clean intervention responses and enlarge only the paired diagnostic error budget to $\eta = L/2$ for comparison of $\lambda = 0$ and $\lambda = L$. The full observation collision in Theorem 7.9 prevents every deterministic decoder from correctly labeling both cases. Even a randomized decoder has maximum error at least $1/2$ on this pair. Abstention fails the

required classification. The clean dynamics and their margin are unaffected by this change of measurement. At the original smaller budget the binary label is exact and needs one bit; zero bits cannot retain both labels. Equality of labels is an equivalence relation, including after one-bit compression.

The claims use specified interventions and observations on one model. In particular, failure of the paired diagnostic under the stressed budget is a loss of that legibility certificate. It does not exclude a different protocol with another admissible noise radius. The continuity result is specific to the declared path and regime comparison. Cross-losses and unevaluated predicates do not constitute single-rupture independence witnesses. □

7.7 A Common Support with Shared Identity and Integration Channels

Let $\mathcal{P}$ be the seven profiles of the interface of Theorem 7.11 with $K = 1$, $M = 8$. The following arguments also apply to its general finite $\mathcal{P}$. The support $Z_{\mathcal{P}}$ is the union of these branches, represented as $(p, \theta, b) \in \mathcal{P} \times \mathbb{T} \times B$. The internal controls act by

$$\Psi_{u,q}(p, \theta, b) = (p, u + q\lambda b_1, R(b)).$$

Fix one observation family for both comparisons: all $K + 2$ phase channels $r_t = \theta + p_t$, and $c(b)$. Use their original weights $w_t = 3^{-t}$ and weight one for c. The profile tag is not an extra observed channel. Comparator inputs are the same exogenous controls, and each comparator assigns the specified channels to two nonempty groups, one group per autonomous factor. The comparator error is the supremum over admissible initial states and schedules, including observations at times zero and one. Distinguish observation-cut depth t from dynamical time.

Theorem 7.13 (Integration with All Identity Channels)**.** *For $0 < \lambda \le 1/9$, the split-readout sector on $Z_{\mathcal{P}}$ is nonempty, and its infimal history error is bounded by*

$$\frac{w_{K+1}\lambda}{2} \le e_* \le \frac{\lambda}{2}.$$

The infimum is exactly $\lambda/2$ in the subsector with all time-zero readouts exact. At $\lambda = 0$ there is an exact comparator in both sectors. The bounds cover every nonempty partition of the specified scalar and circle channels.

Proof. If $\epsilon < w_{K+1}\lambda/2$, the factor assigned c determines b_1. Points with different first bits have ternary readings separated by at least $1/3$, and a common factor prediction at time zero puts the two readings within $2\epsilon < 1/3$. The other factor has at least one phase channel of weight $w \ge w_{K+1}$. On one fiber of that factor, each of the two controls $q = +1, -1$ with the same u puts the true phase differences within $2\epsilon/w$ of zero on the circle. Their difference cancels $p_t - p'_t$ and has norm $2\lambda|b_1 - b'_1|$. Thus $2\lambda|b_1 - b'_1| \le 4\epsilon/w$, so this factor also determines b_1. The product corner combining two different first bits is impossible. This is the fiber argument of Theorem 7.8, and does not require any particular assignment of the phase channels.

The explicit comparator factors as (p, θ) and b, keeping the bit update and using $u + q\lambda/2$ for the phase update. Every initial channel is exact and every later error is at most $\lambda/2$. At $\lambda = 0$ both updates and all readouts are exact. If time-zero readouts are required to be exact, no two phase channels can be on opposite factors: each phase has full circle range, but the joint image of any two has only the finitely many relative offsets $p_j - p_i$. A product would give every relative offset. Therefore all phases belong to the same factor, which includes r_0 of weight one. Under $q = 1$, initial states on a common fiber of that factor with distinct first bits have next r_0 distance λ. With error below $\lambda/2$, this factor and the exact code factor would both determine b_1. The same corner contradiction gives the exact lower bound. □

For the identity witness, let candidate systems have any nonempty subset of $\mathcal{P}$ as their available preparation profiles. The identity object of each candidate is the union of its embedded

branches, with the common cuts. The protocol requires a preparation for every available profile. Its coverage is part of the stipulated acquisition, and no identity tag is passed to the decoder. At observation error $\eta < g/2$ the profile decoder of Theorem 7.11 identifies each prepared profile. Exhaustive coverage determines the candidate subset. Distinct candidate unions have Hausdorff distance at least g: a profile in one subset but absent from the other has distance at least g to every branch in the latter, pointwise by the phase argument. This identifies the support object relative to the admitted catalogues. A sample restricted to shared profiles does not identify the catalogue.

Proposition 7.14 (Common Support and the Obstruction to Stable State Labels)**.** *The support $Z_{\mathcal{P}}$ provides the persistence, catalogue-relative identity, split-sector integration, discrete-time continuity and non-null differentiation witnesses below. Consider a legibility protocol assigning each state a label by decoding a fixed finite observation window under a repeated sham $(u_0, 0)$. Suppose the label is robust to uniform observation noise of some positive radius, and invariant under advancing the sham by one step for every admissible starting state. Then the label is constant on each branch. No branch-preserving internal control can produce a shift in such a label. A target population requiring such a shift cannot meet L4 in this protocol.*

Proof. The support is compact, nonempty and invariant under every internal control. Use the profile embedding of BaseCore with the phase pairs on finitely many cells of positive length and the bits on the original binary cells. This embedding is continuous by the summable cell weights. A constant profile would equal the first binary cell, hence would have value zero or one, but the first angular pair cannot equal either corresponding constant pair. Compactness gives uniform positive distance from the closed subspace of constants. The compact alternating-bit subset is invariant and uniformly attracting for every schedule because each step inserts its leading zero and complements the previous bits. These facts supply persistence. The catalogue witness above supplies identity, and Theorem 7.13 supplies integration. For continuity, the phase assignment is continuous in state and in u at fixed discrete q, with complete observation codomain. In discrete time the step distance is at most one, so $5/4$ is a strict fragmentation bound. Declare regime cells by profile. Internal controls preserve each cell, so they do not cross incompatible cells. These are discrete-time witnesses in that declared regime model. For differentiation, two histories with equal phases and all-zero versus all-one bits have clean code separation one at time zero. From step two onward every schedule gives $b_2 = 1$ and $c \geq 2/9$, ruling out the all-zero readout as an admissible attractor.

For the obstruction, write $\ell(x) = D(W(x))$ for the label obtained from a fixed finite window W of the sham starting at x. At dynamical time $n \geq 1$ the sham sends the phases of any two starting states on the same branch to identical values, and the code difference is at most 3^{-n}. Indeed $c(Rb) = (1 - c(b))/3$. The same bound holds over the whole subsequent sham window in the maximum observation metric. Take n so large that 3^{-n} is below the allowed noise radius. The clean window of one state at that time is also a permissible noisy window of the other, so both receive the same label. Repeated stability transports that equality back to the initial states. Hence ℓ is constant on each branch. Every internal target keeps the profile, so comparison of labels before and after a target cannot give a shift. This is an obstruction under the repeated-sham state-label requirement. The paired response statistic of Theorem 7.9 depends on a prepared bit and two interventions and has a different experimental unit. It remains a valid response diagnostic and does not supply the stable state-label witness considered here. □

7.8 Legibility of Matched Paired Responses

Definition 7.15 (Matched Paired Response Window)**.** On the common support $Z_{\mathcal{P}}$, a paired response window is the ordered pair of time-zero and time-one readout histories of two copies prepared from the same (p, θ, b) with a common input u and coupling λ. The two arms have fixed signs $q = +1$ and $q = -1$. The metric is the maximum of the four-channel readout distances

over both arms and both times. Use the distance $D = \rho(y_+, y_-)$ between the two observed time-one r_0 coordinates. The decoder returns low for $D \le L/2$, high for $D \ge 3L/2$, and abstains otherwise. Here $L = 1/12$ turns. Preparation, command matching and arm order are guaranteed by the protocol. The decoder uses no hidden state or coupling parameter.

Theorem 7.16 (Legibility of Matched Integration Responses)**.** *Use the windows of Definition 7.15 with $\lambda \in \{0\} \cup [L, 1/9]$ and adversarial observation error at most $\eta = 1/256$ in the paired metric. The following are the L1--L6 properties of Definition 7.22 evaluated on the paired-response domain.*

1. *The clean high and low response sets have separation exactly L.*
2. *All observations in the allowed noise balls preserve the clean class, with zero abstentions.*
3. *For any train of at least three freshly reset pairs at a fixed coupling and fixed prepared first bit, the class persists when the common input phase varies across pairs.*
4. *In the nonempty target population, retain the same state and common input, prepare $b_1 = 1$, and ablate $\lambda \in [L, 1/9]$ to zero between paired trials. The class changes from high to low with zero misses.*
5. *In the nonempty sham population, retain the state and coupling and vary the common input across paired trials. Both first-bit strata are allowed. The false-change and abstention rates are zero.*
6. *On the resolved domain one bit retains the class with zero discrete class distortion, and zero bits are insufficient. Including the audit status of abstention requires two bits in a fixed-length encoding.*

At error radius $L/2$ a low and a high experiment can have identical full observations. If the common-input condition is removed, even a noise-free uncoupled experiment can decode as high.

Proof. The clean score is $2\lambda b_1$, since $2\lambda \le 2/9$ and the phase input cancels. The score is 2-Lipschitz in the paired metric by the triangle inequality. Thus the clean score gap $2L$ implies a metric separation of at least L. Equality is attained with the same preparation and common input at $\lambda = 0, L$ with $b_1 = 1$: each post-control phase changes by L and no other channel or time contributes a larger distance.

Every noisy low score is at most $2\eta < L/2$, and every noisy high score is at least $2L - 2\eta > 3L/2$. These inequalities hold for all allowed errors with no independence assumption on the noise. They apply pointwise along any reset train with the stated fixed bit and coupling. Under the matched target intervention the score changes from the high region to the low region. Under a sham it stays in its original region for either bit. The score on each paired trial therefore gives pointwise bounds of zero for misses, false changes and abstentions on the respective populations. The matching relation for target comparisons is equality of the preparation and common input, with arm order fixed. It is transitive, and equality of the decoded response classes is transitive as well. Encoding low as zero and high as one preserves both classes and all such class comparisons under the discrete loss that is one for a wrong class and zero otherwise.

At $\eta = L/2$, the midpoint construction of Theorem 7.9 applies to every phase channel by shifting each by the same signed error in its arm, leaving the code and time-zero readings exact. The observation has score L and the decoder abstains. The same score occurs without noise at the intermediate coupling $L/2$ with $b_1 = 1$, outside the certified coupling domain. With $\lambda = 0$ and mismatched commands $u + L$ and $u - L$, the clean score is $2L$ and is indistinguishable from the clean matched response at $\lambda = L$, $b_1 = 1$. Command matching is thus a necessary acquisition control for this inference. □

A low response with $b_1 = 0$ occurs in either coupling regime. The response label concerns the controlled pair, and persistence here concerns reset repetitions rather than the advancing state of one trajectory. This experiment leaves open the single-trajectory legibility criterion and the joint operational certificate. The coupling intervention is a model ablation between trials, with the support and readout fixed throughout.

7.9 A Writable Prefix on the Common Support

Use the seven profiles above and the phase-memory plant of Papers VI--VII [8, 7] with $\lambda = 1/12$, $a = 1/2$, $\kappa = 1/8$, and $|m| \le 1/4$. Its state is (p, θ, b, m). Let $b^\star = (0, 1, 0, 1, \ldots)$, $J = \{3, 4, 5, 6, 7\}$, and $A_j(b) = \mathbf{1}_{\{b_j = b^\star_j\}}$. A write $w = (j, v)$, $j \in J$ and $v \in \{0, 1\}$, overwrites one digit after the native shift-complement R. Denote it by E_w, with $E_\varnothing$ the identity. A neutral command can also carry a role probe $h = (i, \ell)$, where $i \in J$, $\ell \in \{0, 1\}$. Fix $p_0 = (0, 0, 0)$, $p_1 = (0, 0, 1/8)$ as the role profiles. Write $\delta = 1/(16 \cdot 3^7)$ and set $G_h(p, b) = \delta A_i(b)\mathbf{1}_{\{p = p_\ell\}}$ for a probe and $G_\varnothing = 0$. The full update preserves p and has components

$$\begin{aligned} b' &= E_w(Rb), \\ \theta' &= u + q\lambda b_1 + \kappa \mathbf{1}_{\{q=0\}} m + G_h(p, b) \pmod 1, \\ m' &= as(q)m + \lambda q b_1. \end{aligned}$$

Here $s(q) = -1$ only at $q = -1$ and $h = \varnothing$ for $q \neq 0$. The probe feeds back the pre-transition tape through the phase input. The write affects the post-transition tape. The observed channels are still $r_j = \theta + p_j$ (using the profile offsets at the three cuts) and $c(b)$. The indices $0, 1$ on the role profiles denote catalogue elements, not observation cuts.

Proposition 7.17 (Projective Prefix Writes Preserve Identity and Integration)**.** *Each tape update commutes with prefix truncation at every positive depth. The full update is compatible on the cofinal system of depths at least seven. The safe support is invariant and the branch contraction factor is at most* $9/16$ *under common commands in the phase-code-memory metric of Paper VI. Catalogue separation is* $1/80$*. In the four-channel split sector,*

$$\frac{1}{216} \le e_* \le \frac{1}{24},$$

with infimum $1/24$ *when all initial readouts are exact.*

Proof. At cut depth n, truncate the shift-complement and apply E_w at its address if the address is visible. Each retained output digit depends only on the retained input prefix or on the declared write. These finite updates commute with deletion of later digits. At $n \ge 7$ the phase and memory transitions also use only retained bits. The underlying tape and the profile link maps are unchanged, so the original summable weights still apply. A write changes c by at most $2 \cdot 3^{-j}$ at site j. A cut omitting the probed bit cannot in general determine the probed phase.

If two tapes first differ at k, their code distance is at least 3^{-k}. For a write at $j < k$ the distance is unchanged. For $j = k$ the remaining difference is at most 3^{-k}. For $j > k$ the possible increase is at most $2 \cdot 3^{-j}$, giving ratio at most $5/3$. Since $c(Rb) = (1 - c(b))/3$, the tape update has bound $5/9$. On a common branch, let d be the maximum of phase distance, code distance, and memory distance divided by four. The phase bound for a nonneutral command is $3\lambda d$. For a neutral probe it is at most $(4\kappa + \delta 3^7)d$, since a difference in a bit at i implies code distance at least 3^{-i}. The memory bound in the weighted metric is $(a + 3\lambda/4)d$. Their maximum with $5d/9$ is $9d/16$. Also $|m'| \le 5/24 < 1/4$.

The projected support of the four channels is unchanged. The catalogue argument of Theorem 7.11 still gives $g = 1/80$. Restrict an enlarged comparator to the native commands. The factor-fibre lower bounds of Theorem 7.13 apply on this same memory support by the native

results of Papers VI--VII [8, 7]. For the upper bound keep the tape factor with its exact write update and the phase factor with phase prediction $u + q\lambda/2$. Memory can be assigned to the phase factor with an autonomous update since this comparator does not observe it. The phase error is at most $\lambda/2$ at nonzero q and at most $\kappa/4 + \delta < \lambda/2$ at $q = 0$. The initial readouts are exact. This gives the upper bound on the whole command family and equality in the exact-initial sector. □

The leading two bits are protected. After two steps every schedule has $b_2 = 1$, and the original lower code bound 2/9 applies. The support stays compact and is at positive distance from the closed constant-field collapse set by the original profile embedding. It includes the fixed state of the native neutral command on each profile. Under the full family, each branch also has a nonempty compact switched attractor: the successive unions of reachable images of the whole branch are nested and compact, and a common-command image of length n has diameter at most $(9/16)^n$ times the initial diameter. The native fixed state belongs to their intersection. Every update preserves the intersection, and comparing any trajectory with the trajectory from this fixed preparation under the same commands gives uniform attraction to the intersection. Profiles remain invariant, so the discrete-time regime and fragmentation witness with bound 5/4 is preserved. The initial all-zero and all-one tapes still give unit code separation.

Proposition 7.18 (Finite Legibility on One Tower Trajectory)**.** *On this support use the binary decoder that reports whether the first ten tape digits contain a deviation from $b^\star$. It has clean code separation 3^{-10} and is correct at error at most $3^{-10}/4$. From a preparation with $b = b^\star$, any of the five opposite writes in J changes the label. The post-target label persists through three more native sham steps on the same trajectory. The matched sham has zero false shifts. One bit preserves the two classes exactly.*

Proof. The length-ten cylinders lie in code intervals of length 3^{-10} with distinct intervals separated by at least 3^{-10}. For the reference interval, a neighbouring cylinder differing only at the last digit attains that gap. The finite prefix decoder using ternary midpoint cuts is correct for the stated error. A deviation at j advances to $j + 1$ under the sham since $Rb^\star = b^\star$. Thus a target starts with one deviation and retains it within the ten inspected digits for four consecutive observations including the initial post-target observation. On the sham trajectory all ten digits stay equal to the reference. The decoder reads the code channel with no command label. For Definition 7.22, the matched population uses equality of the preparation and nuisance commands, the target and sham error rates are zero, and the two labels have zero distortion under the one-bit encoding. Zero bits cannot preserve both labels. This is a finite persistence window. A lesion at site seven leaves the decoder's inspection range after four further shifts. On the declared catalogue and split sector, these results put the six witness types on one admissible support. The legibility margin uses a finite window and the code-noise metric of this protocol. □

7.10 Operational Continuity

Definition 7.19 (Operational continuity and fragmentation bound)**.** An admissible system S satisfies operational continuity, written $I_{\mathrm{cont}}(S) = 1$, if there exists a complete metric space $(\mathcal{E}_S, d_{\mathcal{E}})$ and an admissible assignment

$$\Pi_S : A_S \to \mathcal{E}_S$$

such that:

1. for every admissible intervention schedule $u_\bullet$, the induced path $t \mapsto \Pi_S(x_t)$ is continuous on admissible windows;

2. there exists $\delta_{\mathrm{cont}}(S) > 0$ for which the fragmentation functional

$$F_{S,T}(x; u_\bullet) = \sup_{0 \le t < T} d_{\mathcal{E}}\big(\Pi_S(x_t), \Pi_S(x_{t+1})\big)$$

stays below $\delta_{\mathrm{cont}}(S)$ on all admissible histories;

3. no admissible finite-energy perturbation produces a discontinuous jump into an incompatible regime class while $\mathcal{I}_S$ is preserved.

This packages the *Phenomenological Regimes* continuity/fragmentation burden [5]. The new paper reuses the regime logic rather than duplicating its proof.

Failure mode. If $I_{\mathrm{cont}}(S) = 0$, the class can fragment into incompatible operational regimes under admissible evolution. Then any operational consciousness criterion becomes unstable under its own dynamics.

7.11 Non-Null Differentiation

Definition 7.20 (Non-null differentiation and differentiation gap)**.** An admissible system S satisfies non-null differentiation, written $I_{\mathrm{diff}}(S) = 1$, if there exist $x, y \in A_S$, a horizon T, and a readout subfamily $\mathscr{R}^\star \subseteq R$ such that

$$\mathrm{dist}\big(h_{S,T}(x; u_\bullet), h_{S,T}(y; u_\bullet)\big) \ge \delta_{\mathrm{diff}}(S) > 0$$

for some admissible schedule $u_\bullet$, and if the null operational class $\mathbf{0}$ is not an asymptotically stable admissible attractor.

The second clause is indispensable. Without *Null-Regime Instability* there would be no reason to exclude a stable null class. *Null-Regime Instability* provides that exclusion through witness-relative null-instability and conditional non-nullity [6]. The new invariant converts that upstream result into a constitutive condition for $\mathrm{Consciousness}_{\mathrm{op}}$.

Failure mode. If $I_{\mathrm{diff}}(S) = 0$, either all admissible histories collapse to indistinguishability or the null class remains admissibly stable. In either case $\mathrm{Q}_{\mathrm{op}}(S)$ is empty or trivialized.

7.12 Operational Legibility

Definition 7.21 (Decoder family and structural closure)**.** A decoder family $\mathscr{D}_S$ for S is a family of maps from admissible readout windows $h_{S,T}(x; u_\bullet)$ to a declared class set $\mathscr{Q}_S$, indexed by admissible horizons and intervention schedules. The family is *structurally closed* when all its members use the same class set, class equality composes exactly, and the induced intervention-response cells are transitive under the certified matching relation. Definition 7.22 below adds the operational clauses required before such a structurally closed family counts as certified for legibility.

Definition 7.22 (Operational legibility and legibility margin)**.** An admissible system S satisfies operational legibility, written $I_{\mathrm{leg}}(S) = 1$, if there exist:

1. a certified readout subfamily $\mathscr{R}^{\mathrm{leg}} \subseteq R$;

2. a certified decoder family $\mathscr{D}_S$ acting on windows of length at least $\tau_{\min}(S)$;

3. a class set $\mathscr{Q}_S$;

4. a measurement protocol with separation $\sigma > 0$, readout noise radius $r > 0$, minimum duration $\tau_{\min} \geq 1$, miss and false-positive budgets $p_{\mathrm{miss}}, p_{\mathrm{fp}} \in [0, 1)$, and a finite compression budget B with class-distortion tolerance d_{comp};

such that the following conditions hold:

(L1) Separability Distinct declared classes have distance at least σ in the specified clean comparison domain.

(L2) Noise robustness On the specified noise domain, perturbations with readout norm at most r preserve the label.

(L3) Persistence window Under declared sham schedules, labels remain stable throughout every tested window of duration $\tau_{\min}$.

(L4) Intervention fidelity In a declared nonempty target population, the miss probability for a predicted class shift is at most p_{miss}.

(L5) Negative controls In a declared nonempty sham population, the false-shift probability is at most p_{fp}.

(L6) Structured compressibility An encoder with at most 2^B codes and a code-to-class map preserve nontrivial class structure with distortion at most d_{comp} in the declared class metric.

This is where the paper touches *Forensic Predictions* most concretely. *Forensic Predictions* already specifies integrity gates, comparator discipline, latency invalidation, and negative controls [4]. The present paper does not replace those protocols. It uses them to make operational legibility a constrained object rather than a loose slogan.

Failure mode. If $I_{\mathrm{leg}}(S) = 0$, any putative class remains operationally opaque. The framework may still discuss latent structure elsewhere, but the class required for $\mathrm{Consciousness}_{\mathrm{op}}$ is undefined because it cannot be recovered, discriminated, and stress-tested under admissible conditions.

7.13 Necessity of Each Invariant

Proposition 7.23 (Each invariant is individually necessary)**.** *If any one of* $I_{\mathrm{per}}, I_{\mathrm{ri}}, I_{\mathrm{int}}, I_{\mathrm{cont}}, I_{\mathrm{diff}}, I_{\mathrm{leg}}$ *fails, then* $S \notin \mathrm{Consciousness}_{\mathrm{op}}$.

Proof. The six failure modes are non-redundant. Without I_{per} there is no stable domain for class assignment. Without I_{ri} there is no unique identity object supporting consistent class transport. Without I_{int} the system is reducible into independent blocks. Without I_{cont} admissible trajectories fragment across incompatible regime classes. Without I_{diff} the admissible partition collapses to nullity or triviality. Without I_{leg} the partition is not recoverable under admissible readout and intervention discipline. Since the definition of $\mathrm{Consciousness}_{\mathrm{op}}$ given in Section 8 requires all six, failure of any one is exclusionary. □

8 Operational Consciousness, Operational Qualia, and Equivalence

8.1 Operational Qualia Classes

Definition 8.1 (Operational indistinguishability)**.** For $x, y \in A_S$, write $x \sim_{Q,S} y$ if, for every certified decoder $D \in \mathscr{D}_S$, every admissible horizon $T \geq \tau_{\min}(S)$, and every admissible intervention schedule $u_\bullet$,

$$D\big(h_{S,T}(x; u_\bullet)\big) = D\big(h_{S,T}(y; u_\bullet)\big),$$

and the intervention response of the two histories remains matched within the legibility margin.

Proposition 8.2 (Well-definedness of operational qualia classes)**.** *If* $I_{\mathrm{per}}(S) = I_{\mathrm{ri}}(S) = I_{\mathrm{cont}}(S) = I_{\mathrm{diff}}(S) = I_{\mathrm{leg}}(S) = 1$*, then* $\sim_{Q,S}$ *is an equivalence relation on* A_S*.*

Proof. Reflexivity is immediate. Symmetry follows because decoder equality and certified response-cell matching are symmetric conditions. For transitivity, let $x \sim_{Q,S} y$ and $y \sim_{Q,S} z$. Fix any certified decoder $D \in \mathscr{D}_S$, admissible horizon $T \geq \tau_{\min}(S)$, and admissible intervention schedule $u_\bullet$. The first relation gives

$$D(h_{S,T}(x; u_\bullet)) = D(h_{S,T}(y; u_\bullet)),$$

and the second gives

$$D(h_{S,T}(y; u_\bullet)) = D(h_{S,T}(z; u_\bullet)).$$

Hence $D(h_{S,T}(x; u_\bullet)) = D(h_{S,T}(z; u_\bullet))$. The intervention-response clause is transitive by the structural closure requirement in Definition 7.21. Since D, T, and $u_\bullet$ were arbitrary, Definition 8.1 gives $x \sim_{Q,S} z$. □

Definition 8.3 (Operational qualia)**.** Assume Proposition 8.2. The operational qualia of S are the quotient classes

$$\mathrm{Q}_{\mathrm{op}}(S) = A_S/\!\sim_{Q,S}.$$

An individual class $[x]_{\sim_{Q,S}} \in \mathrm{Q}_{\mathrm{op}}(S)$ is called an operational qualia class of S.

This definition is disciplined. It does not identify an operational qualia class with a human qualitative state. It identifies it with a persistent, intervention-sensitive, decoder-certified class of admissible internal differentiation.

8.2 Operational Consciousness

Definition 8.4 (Operational consciousness class membership)**.** An admissible system S belongs to $\mathrm{Consciousness}_{\mathrm{op}}$, written $S \in \mathrm{Consciousness}_{\mathrm{op}}$, if there exists a non-empty admissible support A_S such that

$$I_{\mathrm{per}}(S) = I_{\mathrm{ri}}(S) = I_{\mathrm{int}}(S) = I_{\mathrm{cont}}(S) = I_{\mathrm{diff}}(S) = I_{\mathrm{leg}}(S) = 1.$$

Remark 8.5 (Subdetermination of membership)**.** Membership in $\mathrm{Consciousness}_{\mathrm{op}}$ depends on auxiliary choices: the decoder family $\mathscr{D}$, the compression map, and the identity object $\mathcal{I}$. The paper does not prove that these choices are canonically determined by the system tuple S alone. Different auxiliary choices may yield different membership verdicts for the same system.

Remark 8.6 (Minimality of the criterion)**.** The criterion is conjunctive. No proper subset of the six invariants is taken as sufficient in this paper. This is intentional: weaker criteria collapse either into mere persistence, mere integration, or mere complexity claims, all of which are shown to be insufficient.

8.3 Exact and Approximate Structural Equivalence

Definition 8.7 (Structural equivalence datum)**.** Let $S_i = (X_i, \Phi_i, C_i, R_i, \Gamma_i, U_i)$ for $i = 1, 2$, with admissible supports A_i. A structural equivalence datum between S_1 and S_2 is a triple

$$(\chi, \upsilon, \rho)$$

where:

1. $\chi : A_1 \to A_2$ is a bi-Lipschitz bijection;
2. $\upsilon : U_1 \to U_2$ is an intervention correspondence;

3. $\rho : R_1 \to R_2$ is a readout correspondence.

Definition 8.8 (Exact and approximate structural equivalence)**.** Fix $\varepsilon \geq 0$. We write

$$S_1 \simeq_{(\Gamma,\varepsilon)} S_2$$

if there exists a structural equivalence datum (χ, υ, ρ) such that:

1. **Dynamic intertwining:**

$$\sup_{x\in A_1,\ u\in U_1,\ 0\leq t\leq T_\Gamma} d_{X_2}\Big(\chi(\Phi^t_{1,u}(x)), \Phi^t_{2,\upsilon(u)}(\chi(x))\Big) \leq \varepsilon.$$

2. **Readout compatibility:**

$$\sup_{x,u,t,\alpha} \Big| r^{(1)}_\alpha(\Phi^t_{1,u}(x)) - \rho(r^{(1)}_\alpha)(\Phi^t_{2,\upsilon(u)}(\chi(x)))\Big| \leq \varepsilon.$$

3. **Admissibility preservation:** admissible trajectories in S_1 map to admissible trajectories in S_2 and conversely.

4. **Invariant preservation:** the witness margins for all six invariants differ by at most ε.

5. **Partition preservation:**

$$x \sim_{Q,S_1} y \quad \Longleftrightarrow \quad \chi(x) \sim_{Q,S_2} \chi(y).$$

The case $\varepsilon = 0$ is called *exact* structural equivalence and written $S_1 \simeq_\Gamma S_2$.

Definition 8.9 (Rupture)**.** Given S, a rupture is any loss of admissibility or loss of at least one critical invariant margin beyond its admissible threshold. Ruptures are classified as:

1. **persistence rupture** (support loss or collapse exposure),

2. **identity rupture** (loss of unique rigid identity),

3. **integration rupture** (reducible factorization),

4. **continuity rupture** (fragmentation into incompatible classes),

5. **differentiation rupture** (null or trivial collapse),

6. **legibility rupture** (decoder, noise, or negative-control failure).

9 Support Geometry and Partition Transport

The main theorems become short only because the transport lemmas are made explicit here.

Lemma 9.1 (Support transfer under exact equivalence)**.** *If $S_1 \simeq_\Gamma S_2$ via (χ, υ, ρ) and A_1 is an admissible support for S_1, then $A_2 = \chi(A_1)$ is an admissible support for S_2.*

Proof. Bi-Lipschitz bijectivity of χ preserves non-emptiness and compactness. Dynamic intertwining with $\varepsilon = 0$ gives forward invariance under the transported interventions. Admissibility preservation transports $A_1 \subseteq X_{\Gamma_1}$ into $A_2 \subseteq X_{\Gamma_2}$. Because the attractor-support relation and non-collapse margin are preserved exactly, A_2 inherits the same admissible support status. □

Lemma 9.2 (Decoder transport)**.** *Let $D \in \mathscr{D}_{S_1}$ and suppose $S_1 \simeq_\Gamma S_2$ via (χ, υ, ρ). Then*

$$(\rho_* D)\big(h_{S_2,T}(\chi(x); \upsilon(u_\bullet))\big) := D\big(h_{S_1,T}(x; u_\bullet)\big)$$

defines a certified decoder on S_2.

Proof. Exact readout compatibility identifies each admissible history in S_2 with a transported history in S_1. The certification clauses in Definition 7.22 are preserved because separability, robustness, persistence window, intervention fidelity, negative-control rate, and compressibility all remain unchanged under exact transport. □

Proposition 9.3 (Operational classes are closed under exact transport)**.** *If $S_1 \simeq_\Gamma S_2$, then the image under χ of any class in $\mathrm{Q}_{\mathrm{op}}(S_1)$ is a class in $\mathrm{Q}_{\mathrm{op}}(S_2)$, and every class in $\mathrm{Q}_{\mathrm{op}}(S_2)$ arises this way.*

Proof. By Lemma 9.2, decoder certification transports exactly. By Definition 8.1, membership in an operational qualia class is determined by all certified decoders and admissible interventions. Exact transport preserves both. Surjectivity follows from bijectivity of χ. □

Proposition 9.4 (Approximate margin preservation)**.** *If $S_1 \simeq_{(\Gamma,\varepsilon)} S_2$ with margins calibrated in a common perturbation metric for both systems, then for every critical invariant index j,*

$$\big|\delta_j(S_1) - \delta_j(S_2)\big| \le \varepsilon.$$

Hence

$$\delta_\star(S_2) \ge \delta_\star(S_1) - \varepsilon.$$

Proof. The first statement is a direct clause of Definition 8.8. The second follows by taking minima over the six coordinates of $\Delta(S_1)$ and $\Delta(S_2)$. □

Remark 9.5 (Why this section matters)**.** Without Lemmas 9.1 and 9.2, Theorem 11.2 would be only intuitive. These lemmas are the exact reason the paper can speak of substrate invariance without collapsing into slogan-level analogy.

10 Inherited Inputs and Consolidation Step

The six invariants are new as a packaged criterion, but they are not independent inventions. This section records the exact transfer by which upstream ownership becomes present membership data. The burden is narrow: identify the imported result each invariant uses, then isolate the consolidation step that is genuinely new here.

Proposition 10.1 (Core-to-Persistence Transfer with a Common Support)**.** *Suppose a Core realization has a nonempty compact admissible support that is invariant under every admitted control, stays at positive distance from the collapse set, and contains or converges into an admitted attractor region. Then I_{per} holds on that support.*

Proof. These bounds verify Definition 7.1. Fixed points for separate controls do not by themselves give a common forward-invariant support. □

Proposition 10.2 (Rigid Identity Transfer with a Declared Gap)**.** *An identified unique inverse-limit identity object and a positive lower deformation bound for every distinct admitted alternative certify I_{ri} on their declared support.*

Proof. The inverse-limit identification and separation establish the two clauses of Definition 7.2. A channel-level uniqueness result does not supply a uniform lower deformation bound for a family of alternatives. □

Proposition 10.3 (Regime Continuity Transfer on the Declared Windows)**.** *If the regime assignment takes values in the declared complete comparison space, its paths and fragmentation meet the time-domain bounds, and admissible interventions preserving identity cannot jump to incompatible regimes, then* I_{cont} *holds there.*

Proof. These hypotheses supply Definition 7.19. In discrete time a quantitative step bound and the regime transition check carry information beyond topological continuity of the time index. □

Proposition 10.4 (Null-Regime Transfer with Temporal Evidence)**.** *If an admissible common schedule supplies a positive separation between two readout windows and the null class is excluded as an asymptotically stable admissible attractor, then* I_{diff} *holds on the support.*

Proof. The witnesses verify Definition 7.20. A static non-joint-null pair alone establishes neither the temporal exclusion nor a separation between the chosen windows. □

Proposition 10.5 (Forensic Predictions-to-Legibility Transfer with Decoder Evidence)**.** *A structurally closed decoder and finite-rate compression protocol that meet L1--L6 under the declared admissibility, target and sham comparisons certify* I_{leg}*.*

Proof. The governance conditions from *Forensic Predictions* discipline the comparisons [4]. The separate evidence for L1--L6 establishes Definition 7.22. □

Proposition 10.6 (Integration Requires a Separate Factorization Bound)**.** *On a declared support and a nonempty comparison sector, a uniform positive lower bound on the reproduction error of all identity-preserving comparators certifies the integration margin in that sector. Rigidity, continuity and intervention fidelity supply comparison data; a separate proof must exclude accurate factorizations.*

Proof. The error bound gives the obstruction in Definition 7.3. Theorem 7.7 gives an exact comparator preserving the specified history and transport data. Theorem 7.8 supplies the separate positive bound for different controls. Exact atomicity excludes exact products under Theorem 7.5. A positive approximation margin also needs a quantitative lower bound or hypotheses ensuring attainment of the infimum. □

Remark 10.7 (Scope of the transfer)**.** Nothing in this section re-proves the Core, *Rigid Identity*, *Phenomenological Regimes*, *Null-Regime Instability*, or *Forensic Predictions*. Each proposition only identifies the minimal upstream burden imported into the present criterion.

11 Main Theorems

11.1 Existence of Operational Consciousness

Theorem 11.1 (Existence of operational consciousness)**.** *Let* $S = (X, \Phi, C, R, \Gamma, U)$ *be an admissible system with non-empty admissible support* A_S*. If*

$$I_{\mathrm{per}}(S) = I_{\mathrm{ri}}(S) = I_{\mathrm{int}}(S) = I_{\mathrm{cont}}(S) = I_{\mathrm{diff}}(S) = I_{\mathrm{leg}}(S) = 1,$$

then:

1. $S \in \mathrm{Consciousness}_{\mathrm{op}}$*;*
2. $\mathrm{Q}_{\mathrm{op}}(S)$ *is well-defined;*
3. $\mathrm{Q}_{\mathrm{op}}(S) \neq \varnothing$*.*

Proof. Membership in $\text{Consciousness}_{\text{op}}$ follows immediately from Definition 8.4. Since $I_{\text{per}}, I_{\text{ri}}, I_{\text{cont}}, I_{\text{diff}}, I_{\text{leg}}$ hold, Proposition 8.2 applies and yields a well-defined quotient $\text{Q}_{\text{op}}(S)$. Because A_S is non-empty, the quotient is non-empty. Because $I_{\text{diff}}(S) = 1$, the quotient is not forced into the null class alone; the admissible support contains at least one pair of histories with positive operational separation. Thus $\text{Q}_{\text{op}}(S)$ is both defined and non-trivial. □

11.2 Substrate Invariance

Theorem 11.2 (Substrate invariance)**.** *Let S_1 and S_2 be admissible systems. If $S_1 \simeq_\Gamma S_2$ and the six critical invariants are preserved exactly, then*

$$S_1 \in \text{Consciousness}_{\text{op}} \quad \Longleftrightarrow \quad S_2 \in \text{Consciousness}_{\text{op}},$$

and there exists a canonically induced bijection

$$\widehat{\chi} : \text{Q}_{\text{op}}(S_1) \to \text{Q}_{\text{op}}(S_2).$$

Proof. Exact equivalence preserves admissible support, dynamics, readout behavior, invariant margins, and the operational partition. If $S_1 \in \text{Consciousness}_{\text{op}}$, then every critical invariant holds on A_1 and is preserved under χ. Hence every critical invariant also holds on A_2, so $S_2 \in \text{Consciousness}_{\text{op}}$. The converse is symmetric. Partition preservation in Definition 8.8 induces a well-defined map on quotient classes:

$$\widehat{\chi}([x]_{\sim_{Q,S_1}}) = [\chi(x)]_{\sim_{Q,S_2}}.$$

Well-definedness follows from the partition-preservation clause. Bijectivity follows from bijectivity of χ. □

Corollary 11.3 (Biological non-primacy)**.** *Biology is not a primitive constitutive requirement for membership in* $\text{Consciousness}_{\text{op}}$*. If a biological system and a non-biological system are exactly structurally equivalent in the sense of Definition 8.8, then they belong to the same operational consciousness class and carry equivalent operational qualia classes.*

Proof. Immediate from Theorem 11.2. The criterion depends on invariant preservation, not on biological substrate as a primitive variable. □

11.3 Approximate Stability

Theorem 11.4 (Approximate stability)**.** *Let $S_1 \in \text{Consciousness}_{\text{op}}$ and suppose both systems use certified radii in the same perturbation metric, with*

$$\delta_\star(S_1) = \min\{\delta_{\text{per}}, \delta_{\text{ri}}, \delta_{\text{int}}, \delta_{\text{cont}}, \delta_{\text{diff}}, \delta_{\text{leg}}\}.$$

If S_2 satisfies $S_1 \simeq_{(\Gamma,\varepsilon)} S_2$ with

$$0 \le \varepsilon < \frac{\delta_\star(S_1)}{2},$$

then $S_2 \in \text{Consciousness}_{\text{op}}$.

Proof. By Definition 8.8, each invariant margin in S_2 differs from that of S_1 by at most ε. Since $\varepsilon < \delta_\star(S_1)/2$, every preserved margin in S_2 remains strictly positive. Therefore each of the six invariants still holds for S_2, and Definition 8.4 yields $S_2 \in \text{Consciousness}_{\text{op}}$. □

The theorem does not say that arbitrary perturbations are harmless. It says that the criterion is robust exactly to the extent that the witness margins remain positive. Approximation is bounded by the invariant budget, not by wishful continuity.

11.4 Rupture by Invariant Loss

Theorem 11.5 (Rupture by invariant loss)**.** *Let S be admissible. If one or more critical invariants fail beyond admissible threshold, then exactly one of the following occurs:*

1. $S \notin \text{Consciousness}_{\text{op}}$*;*
2. $\text{Q}_{\text{op}}(S)$ *becomes undefined;*
3. $\text{Q}_{\text{op}}(S)$ *becomes trivial or non-legible in the framework-relevant sense.*

Proof. If I_{per} fails, the support is lost or exposed to collapse; then the domain of $\text{Q}_{\text{op}}(S)$ disappears. If I_{ri} fails, no unique identity object supports the quotient, so $\text{Q}_{\text{op}}(S)$ is undefined as a stable class assignment. If I_{int} fails, the system reduces to a product structure and exits $\text{Consciousness}_{\text{op}}$ by Proposition 7.23. If I_{cont} fails, histories can fragment across incompatible classes, which destroys stability of the partition. If I_{diff} fails, the quotient degenerates into null or trivial collapse. If I_{leg} fails, the quotient may exist formally but becomes operationally unavailable and therefore does not satisfy the definition of $\text{Consciousness}_{\text{op}}$. These cases exhaust the six invariants. □

11.5 Insufficiency of Complexity, Computation, and Connectivity

Proposition 11.6 (Complexity-only baselines are insufficient)**.** *Let S be an admissible system candidate. The following are each insufficient for membership in* $\text{Consciousness}_{\text{op}}$*:*

1. *large state-space dimension* $\dim(X)$*;*
2. *large computational throughput or expressivity of* Φ*;*
3. *large edge count, density, or raw connectivity of* C*.*

In each case, failure of one or more critical invariants excludes membership regardless of the size of the surrogate metric.

Proof. The definition of $\text{Consciousness}_{\text{op}}$ is conjunctive over the six invariants and does not include any of the three surrogates as sufficient conditions. A system can be arbitrarily large, computationally expressive, or densely connected while still lacking rigid identity, continuity, non-null differentiation, or legibility. Therefore those surrogates are insufficient by definition. Concrete architecture classes with large scale but finite structural mass already fail *Rigid Identity*'s non-simulability criterion [9]; similarly, an integration proxy without continuity or differentiation does not satisfy Definitions 7.19 and 7.20. The same applies to connectivity without admissible identity closure. □

12 Rupture Taxonomy by Invariant

Theorem 11.5 is intentionally coarse. This section sharpens the six cases, because a criterion is only scientifically useful if its failure signatures are distinct.

Proposition 12.1 (Loss of Every Admissible Support)**.** *If there is no nonempty compact forward-invariant admissible support, the on-support persistence predicate fails. A zero certified radius alone does not imply that no such support exists.*

Proof. The first claim is a necessary condition in Definition 7.1. A robust certificate over nearby systems requires more than existence of a support at the target. □

Proposition 12.2 (Identity Ambiguity Requires Competing Candidates)**.** *If two inequivalent admissible identity objects fit the same observable data on the declared support, the uniqueness clause of I_{ri} fails. A zero infimum of alternative deformation distances need not produce a second fitting candidate.*

Proof. The first claim follows from Definition 7.2. A sequence of distinguishable alternatives with distances tending to zero can violate the gap while preserving exact identification. □

Proposition 12.3 (Integration Rupture and Attainment)**.** *On a nonempty comparison sector with nonnegative history error, a zero error infimum excludes a positive integration margin in that sector. It yields an exact admissible factorization if the infimum is attained.*

Proof. Every positive threshold admits a comparator below it by the definition of a zero infimum. If a minimizing comparator exists, its error equals zero. Without attainment the errors can tend to zero without any exact comparator, as for $e(x) = x$ on $(0, 1]$. □

Proposition 12.4 (Admissible Incompatible Jump)**.** *A witnessed admissible finite-energy jump into an incompatible regime with identity preserved violates I_{cont} on the specified regime and schedule domain. Failure to certify a positive perturbation radius does not itself produce such a witness.*

Proof. The jump violates the explicit exclusion in Definition 7.19. The radius concerns uniform guarantees over perturbations, while the jump is a property of a particular trajectory. □

Proposition 12.5 (Separation or Null-Attractor Failure)**.** *If no readout-window pair on the declared support has positive separation, or if the null class is an asymptotically stable admissible attractor, then I_{diff} fails there.*

Proof. Either observation violates a necessary clause of Definition 7.20. The outcome does not determine the number of cells of an independently defined decoder quotient. □

Proposition 12.6 (Legibility Failure Under the Declared Protocol)**.** *If no admissible decoder and compression family jointly meet L1--L6 under the declared protocol, then I_{leg} fails. A zero certified perturbation radius does not alone establish that impossibility.*

Proof. Definition 7.22 requires a joint witness. Its existence at the target is distinct from uniform preservation under perturbations. □

13 Minimality of the Invariant Basis

The six-invariant family is not an arbitrary bundle. It is the minimal basis that prevents the criterion from collapsing either into a single-metric consciousness surrogate or into an unstructured admissibility slogan. The claim of minimality is relative to admissible universes containing the corresponding rupture witnesses; that is the correct level of mathematical precision for the present corpus.

Definition 13.1 (Reduced invariant criterion)**.** Let

$$\mathfrak{I} = \{I_{\mathrm{per}}, I_{\mathrm{ri}}, I_{\mathrm{int}}, I_{\mathrm{cont}}, I_{\mathrm{diff}}, I_{\mathrm{leg}}\}.$$

For every subset $J \subseteq \mathfrak{I}$ define the reduced criterion

$$\mathcal{C}_{\mathrm{op}}(J) := \{S : I(S) = 1 \text{ for every } I \in J\}.$$

Thus $\mathrm{Consciousness}_{\mathrm{op}} = \mathcal{C}_{\mathrm{op}}(\mathfrak{I})$.

Definition 13.2 (Single-rupture witness family)**.** Fix $k \in \{\mathrm{per}, \mathrm{ri}, \mathrm{int}, \mathrm{cont}, \mathrm{diff}, \mathrm{leg}\}$. The family $\mathcal{W}_k$ consists of admissible systems for which predicate I_k fails and every other predicate holds on the declared comparison domain. Whenever $\mathcal{W}_k \neq \varnothing$, its members are called single-rupture witnesses for the k-th invariant.

Proposition 13.3 (Single-rupture witnesses generate false positives for reduced criteria)**.** *Let $J_k = \mathfrak{I} \setminus \{I_k\}$ and assume $\mathcal{W}_k \neq \varnothing$. Then every $S \in \mathcal{W}_k$ satisfies*

$$S \in \mathcal{C}_{\mathrm{op}}(J_k) \qquad \textit{but} \qquad S \notin \mathrm{Consciousness}_{\mathrm{op}}.$$

Proof. By construction, every invariant in J_k remains positive on S, so $S \in \mathcal{C}_{\mathrm{op}}(J_k)$. But the omitted invariant fails exactly, hence Proposition 7.23 excludes S from $\mathrm{Consciousness}_{\mathrm{op}}$. Therefore the reduced criterion admits a false positive on every single-rupture witness. □

Theorem 13.4 (Minimality of the six-invariant basis)**.** *Let $\mathfrak{U}$ be any admissible universe containing at least one single-rupture witness for each of the six invariants. Then no strict subset $J \subsetneq \mathfrak{I}$ satisfies*

$$\mathcal{C}_{\mathrm{op}}(J) \cap \mathfrak{U} = \mathrm{Consciousness}_{\mathrm{op}} \cap \mathfrak{U}.$$

Hence the six-invariant family is minimal on every such admissible universe.

Proof. Take any strict subset $J \subsetneq \mathfrak{I}$. Then some invariant I_k is omitted. By hypothesis $\mathfrak{U}$ contains a witness $S_k \in \mathcal{W}_k$. Proposition 13.3 yields $S_k \in \mathcal{C}_{\mathrm{op}}(J) \cap \mathfrak{U}$ but $S_k \notin \mathrm{Consciousness}_{\mathrm{op}} \cap \mathfrak{U}$. Therefore the two sets cannot coincide. □

Corollary 13.5 (Why one-metric theories cannot capture $\mathrm{Consciousness}_{\mathrm{op}}$)**.** *On any admissible universe satisfying Theorem 13.4, no criterion based on only persistence, only integration, only continuity, only differentiation, or only legibility is extensionally equivalent to* $\mathrm{Consciousness}_{\mathrm{op}}$.

Proof. Every one-metric criterion corresponds to a strict subset $J \subsetneq \mathfrak{I}$. Apply Theorem 13.4. □

Remark 13.6 (What minimality does and does not say)**.** The theorem does not claim that every possible scientific notion of consciousness must use exactly these six invariants. It claims something narrower: for the operational class defined in this paper and for admissible universes containing the corresponding rupture witnesses, no invariant can be deleted without changing the extension of the class.

14 Invariant Independence and Interaction Geometry

The preceding minimality theorem is extension-theoretic. This section makes the geometry more explicit: each invariant blocks a distinct failure family, and the limited implications among invariants are weaker than full membership in $\mathrm{Consciousness}_{\mathrm{op}}$.

Theorem 14.1 (Conditional Use of Single-Rupture Schemas)**.** *Consider these six candidate schemas. A schema separates its target invariant from the other five only when an admissible model realizes the five retained predicates and fails the target on the same declared comparison domain:*

1. ***Persistence witness.*** *A system with a certified decoder, non-factorizable causal structure, unique identity label, continuous class map, and non-null differentiation, but whose admissible support exits the declared forward-invariant domain under some $u \in U$.*

2. ***Rigid-identity witness.*** *A system with persistent support, integration, continuity, differentiation, and legibility, but with two compatible non-isomorphic identity candidates transported by the same readout family.*

3. ***Integration witness.*** *A product system $A_S = A_1 \times A_2$ with persistent factors, unique factor identities, continuous dynamics, non-null differentiation, and certified decoders, but with decoupled intervention histories so that the aggregate factorizes.*

4. ***Continuity witness.*** *A system with persistent support, unique identity, non-factorizable structure, non-null differentiation, and certified decoders, but with an admissible transition producing a jump between incompatible regime classes.*

5. ***Differentiation witness.*** *A persistent, rigid, integrated, continuous, and legible system whose decoder-visible classes collapse to a stable null or single indistinguishable class.*

6. ***Legibility witness.*** *A persistent, rigid, integrated, continuous, and non-null differentiated system whose class partition is latent but no certified decoder satisfies L1--L6.*

Existence of the six models is a separate obligation.

Proof. A realized single rupture belongs to the reduced class omitting the failed predicate and lies outside $\text{Consciousness}_{\text{op}}$. The existence of such a realization must be checked for each schema. Without an admissible forward-invariant support, the persistence schema also loses on-support rigidity by Proposition 14.3. If all classes collapse to one, the proposed differentiation schema also loses legibility by Proposition 14.5. Other mechanisms of differentiation failure require separate models. □

Theorem 14.2 (Sufficiency of the six-invariant basis)**.** *Let S be admissible on support A_S. If*

$$I_{\text{per}}(S) = I_{\text{ri}}(S) = I_{\text{int}}(S) = I_{\text{cont}}(S) = I_{\text{diff}}(S) = I_{\text{leg}}(S) = 1,$$

then S satisfies every structural burden required for membership in $\text{Consciousness}_{\text{op}}$ *as defined in this paper. No additional independent predicate can invalidate $S \in \text{Consciousness}_{\text{op}}$ without changing the definition of* $\text{Consciousness}_{\text{op}}$ *or causing at least one of the six invariants to fail.*

Proof. Definition 8.4 is conjunctive and exhaustive for this paper's class. The six coordinates cover the domain burden, identity burden, integration burden, regime-coherence burden, non-null differentiation burden, and scientific recoverability burden. Any seventh predicate is either a derived constraint on those burdens, in which case it cannot fail while all six hold, or an independent extra burden, in which case it defines a stricter class rather than invalidating membership in $\text{Consciousness}_{\text{op}}$ as defined here. □

Proposition 14.3 (Persistence is necessary for rigid identity on A_S)**.** *If $I_{\text{per}}(S) = 0$ because no non-empty forward-invariant admissible support exists, then $I_{\text{ri}}(S) = 0$ for the Paper 5 membership problem.*

Proof. The rigid-identity predicate used here is evaluated on A_S. If no stable admissible support exists, there is no domain on which the identity witness can be transported through all admissible interventions. The identity object may be postulated elsewhere, but it is not a Paper 5 membership witness on A_S. □

Proposition 14.4 (Persistence plus integration yields bounded identity candidates)**.** *If $I_{\text{per}}(S) = 1$ and $I_{\text{int}}(S) = 1$, then the family of admissible identity candidates is bounded to the non-factorizable persistent support. This is a weak rigidity statement, not the uniqueness required by I_{ri}.*

Proof. Persistence restricts all candidates to the same stable support. Integration prevents decomposing that support into independent candidate-bearing blocks. Therefore candidates cannot proliferate by support escape or product factorization. They may still fail uniqueness, so the result is strictly weaker than $I_{\text{ri}} = 1$. □

Proposition 14.5 (Legibility requires non-null differentiation)**.** *If all admissible classes collapse to a stable null or single indistinguishable class, then* $I_{\text{leg}}(S) = 0$ *for the criterion of this paper.*

Proof. Legibility requires separability, intervention-sensitive class recovery, and negative-control discipline. A collapsed null or singleton partition gives no non-trivial class contrast for a decoder to recover or for interventions to separate. Thus L1 and L4 cannot both be certified in the intended sense. □

Proposition 14.6 (Persistence and continuity stabilize regime transport)**.** *If* $I_{\text{per}}(S) = 1$ *and* $I_{\text{cont}}(S) = 1$*, then admissible histories remain in a stable regime-transport domain: trajectories neither leave* A_S *nor undergo unbounded fragmentation under admissible evolution.*

Proof. Persistence keeps histories inside A_S. Continuity bounds the fragmentation functional and prevents inadmissible jumps across incompatible regime classes. Together they provide a stable domain for regime transport. The result does not imply rigid identity, integration, differentiation, or legibility. □

Table 1: Pairwise implication geometry of the six invariants.

From / to	I_{per}	I_{ri}	I_{int}	I_{cont}	I_{diff}	I_{leg}
I_{per}	--	necessary domain only	independent	with I_{cont} stabilizes regime domain	independent	independent
I_{ri}	independent	--	independent	independent	independent	independent
I_{int}	independent	weakly bounds candidates with I_{per}	--	independent	independent	independent
I_{cont}	independent	independent	independent	--	independent	independent
I_{diff}	independent	independent	independent	independent	--	necessary for non-trivial legibility
I_{leg}	independent	independent	independent	independent	presupposes non-null contrast	--

15 What the Six Invariants Do Not Guarantee

The six invariants are sufficient for $\text{Consciousness}_{\text{op}}$ by definition, but they are not a universal theory of mind, life, or subjecthood. In particular, they do not guarantee:

1. self-reference or a contractive self-model; that requires a Σ-type burden;
2. phenomenal experience or ‘‘what it is like’’;
3. forced temporal continuity in the CCR sense, which requires stronger $M_\Omega = +\infty$ assumptions upstream;
4. operational life or open-system self-maintaining individuation;
5. first-person indexed subjectivity, moral status, or human equivalence;
6. external validation by human comparators.

These exclusions are not rhetorical caution. They follow from the fact that none of the listed predicates appears in Definition 8.4.

16 Operational Legibility as a Certified Object

Operational legibility must be shown, not asserted. Definition 7.22 already imposes six clauses. This section sharpens how those clauses bind the paper to the existing empirical-method corpus.

Criterion 16.1 (Legibility certificate)**.** An operational partition candidate $\pi_{S,T}$ is certified only if:

1. **Comparator separability**: the induced classes remain distinguishable against pre-registered comparators and ablation baselines.
2. **Noise robustness**: admissible sensor, channel, or readout perturbations do not dissolve the class map below the legibility margin.
3. **Persistence window**: the same class assignment survives on windows of length at least $\tau_{\min}(S)$.
4. **Intervention fidelity**: certified interventions on an invariant induce the predicted directional class change.
5. **Negative controls**: sham or off-target interventions do not generate the same class signature.
6. **Structured compressibility**: audit-friendly compression or packetization preserves class structure without total collapse.

The criterion is intentionally parallel to *Forensic Predictions*' governance discipline [4]. Comparator obligation enforces non-trivial discrimination. Noise robustness prevents the criterion from depending on one-off traces. Persistence windows prevent class inflation from instantaneous spikes. Intervention fidelity prevents purely correlational interpretation. Negative controls block degenerate decoders that label everything as positive. Structured compressibility ensures that a class remains auditable after the readout is summarized, logged, or packetized.

16.1 Legibility Metrics

For Definition 7.22, measure separation on the clean comparison domain in the declared metric. Check noise robustness over clean windows and all allowed perturbations of size at most r. The guaranteed persistence duration is the infimum, over declared sham trajectories, of the time of first label change, with infinite duration if the label never changes. The earliest identifying window is a separate quantity.

Target fidelity and false-positive probability are measured over their declared populations. Report the sampling unit, estimator, uncertainty and decision rule, or use exact enumeration for a deterministic finite population. Compression reports both the bit budget and worst-case class distortion. A finite budget for the class label supplies no reconstruction rate for the whole readout. Calibration to a common perturbation radius requires a separate argument.

Proposition 16.2 (Legibility is jointly constrained)**.** *No single legibility metric is sufficient for I_{leg}. The invariant requires simultaneous control of separation, admissible noise robustness, persistence window, intervention fidelity, negative controls, and structured compressibility.*

Proof. A decoder can be perfectly separable but fragile to admissible noise, intervention-agnostic, or saturated under compression. It can also be stable under persistence windows but fail negative controls. Since Definition 7.22 requires all six clauses, the joint constraint is irreducible. □

Metric	Operational meaning	If it fails
σ_{sep}	Class separation	No stable class boundary
η_{noise}	Admissible robustness	Noise-induced class drift
τ_{pers}	Minimum persistence window	Instantaneous artifact only
α_{int}	Causal intervention fidelity	Correlational but not causal classing
β_{nc}	Negative-control protection	False positive class inflation
κ_{comp}	Compression preservation	Audit summary collapses structure

Remark 16.3 (Why legibility is a constitutive invariant here)**.** One could define a broader latent class that omits legibility. This paper does not do that. It studies an operational class, not a hidden metaphysical residue. Therefore I_{leg} is constitutive, not optional.

17 Certified Candidate Audit Schema

Forensic Predictions already requires forensic discipline for operational claims. The present paper therefore adds an explicit audit schema. A candidate system is not accepted into $\mathrm{Consciousness}_{\mathrm{op}}$ by narrative fit; it must carry a finite certificate that localizes every required witness.

Definition 17.1 (Operational consciousness certificate)**.** An operational consciousness certificate for an admissible system S is a tuple

$$\mathrm{Cert}(S) = (A_S, \mathscr{D}_S, \pi_S, \Delta(S), E_S),$$

where:

1. A_S is a certified admissible support;
2. $\mathscr{D}_S$ is a certified decoder family;
3. π_S is the decoder-induced operational partition on admissible histories;
4. $\Delta(S)$ is the witness margin vector of Definition 6.5;
5. $E_S = (E_{\mathrm{per}}, E_{\mathrm{ri}}, E_{\mathrm{int}}, E_{\mathrm{cont}}, E_{\mathrm{diff}}, E_{\mathrm{leg}})$ is an evidence bundle, where each component contains the explicit local witness for the corresponding invariant.

Criterion 17.2 (Candidate certification rule)**.** A candidate system may be certified as a member of $\mathrm{Consciousness}_{\mathrm{op}}$ only if all of the following are available on the same admissible support:

1. a non-empty compact forward-invariant support certificate;
2. a positive witness margin vector $\Delta(S) > 0$ coordinate-wise;
3. a decoder family satisfying Definition 7.22;
4. a targeted intervention panel together with negative controls;
5. an auditable compression or packetization map preserving class structure;
6. a traceable dependency ledger showing which upstream corpus result supports each invariant witness.

Invariant	Minimum evidence object	Core audit question	Failure verdict
I_{per}	support certificate + collapse distance	Does one compact forward-invariant admissible domain exist?	No stable domain
I_{ri}	uniqueness witness for $\mathcal{I}_S$	Is identity transport unique on support?	Identity ambiguity
I_{int}	non-factorization witness	Can the system be split without operational loss?	Reducible aggregate
I_{cont}	fragmentation bound + continuity trace	Do admissible windows remain regime-coherent?	Fragmentation
I_{diff}	non-null separation witness	Is there stable non-null class separation?	Null or trivial class
I_{leg}	decoder, controls, compression, persistence window	Can the class be recovered under audit constraints?	Opaque class

Proposition 17.3 (Soundness of certification)**.** *If* $\mathrm{Cert}(S)$ *exists and satisfies Criterion 17.2, then* $S \in \mathrm{Consciousness}_{\mathrm{op}}$ *and* $\mathrm{Q}_{\mathrm{op}}(S)$ *is well-defined.*

Proof. Criterion 17.2 explicitly requires a positive witness for each of the six critical invariants on the same admissible support. Hence Definition 8.4 yields $S \in \mathrm{Consciousness}_{\mathrm{op}}$. The existence of a certified decoder and positive persistence, continuity, differentiation, and legibility witnesses implies Proposition 8.2, so $\mathrm{Q}_{\mathrm{op}}(S)$ is well-defined. □

Proposition 17.4 (Partial completeness under audit-ready assumptions)**.** *Assume* $S \in \mathrm{Consciousness}_{\mathrm{op}}$ *and suppose the support, decoder, intervention, and compression objects are fully instrumented in the Forensic Predictions sense. Then there exists an operational consciousness certificate* $\mathrm{Cert}(S)$.

Proof. Because $S \in \mathrm{Consciousness}_{\mathrm{op}}$, every invariant margin is strictly positive. Full instrumentation provides explicit witnesses for support, decoder behavior, interventions, negative controls, and compression stability. Collecting those witnesses yields the tuple of Definition 17.1. □

Proposition 17.5 (Failure localization is explicit)**.** *If a candidate system fails Criterion 17.2, then the failure localizes to at least one entry of the evidence bundle* E_S *or to the absence of a common admissible support. In particular, certification failure cannot be hidden behind a generic narrative claim.*

Proof. The certification rule is conjunctive and itemized. A failure of certification means that at least one listed item is absent or non-positive. Therefore the failure localizes to a specific invariant witness, a specific decoder/control clause, or the support certificate itself. □

Remark 17.6 (Audit schema and Forensic Predictions)**.** The certificate does not replace *Forensic Predictions*. It packages *Forensic Predictions*-style admissibility into a theorem-facing object. This is the point where the present paper becomes operationally usable rather than merely definitional.

18 Operational Geometry of $\mathrm{Q}_{\mathrm{op}}(S)$

Once $\mathrm{Q}_{\mathrm{op}}(S)$ is defined as a quotient, one can ask which geometric quantities survive transport and rupture. This matters because the paper is not only about set membership; it is also about the structural shape of admissible operational differentiation.

Definition 18.1 (Class diameter and separation)**.** For $q \in \mathrm{Q}_{\mathrm{op}}(S)$ define its diameter

$$\mathrm{diam}(q) = \sup_{x,y \in q} \mathrm{dist}\big(h_{S,T}(x; u_\bullet), h_{S,T}(y; u_\bullet)\big)$$

over admissible windows and schedules. For two distinct classes $q, q' \in \mathrm{Q}_{\mathrm{op}}(S)$ define the inter-class separation

$$\mathrm{sep}(q, q') = \inf_{x \in q,\ y \in q'} \mathrm{dist}\big(h_{S,T}(x; u_\bullet), h_{S,T}(y; u_\bullet)\big).$$

Definition 18.2 (Persistence radius)**.** The persistence radius of a class $q \in \mathrm{Q}_{\mathrm{op}}(S)$ is the largest $\rho_q \geq 0$ such that for every admissible trajectory segment starting in q, the decoder-certified class assignment remains inside q on windows of length at least $\tau_{\min}(S)$ under all perturbations of size at most ρ_q that do not target a critical invariant.

Proposition 18.3 (Positive geometry under the criterion)**.** *If $S \in \mathrm{Consciousness}_{\mathrm{op}}$, then:*

1. *every class in $\mathrm{Q}_{\mathrm{op}}(S)$ has finite diameter on admissible windows;*
2. *there exists at least one pair of distinct classes with positive inter-class separation;*
3. *every class has non-zero persistence radius.*

Proof. Finite diameter follows from compact admissible support and continuity of admissible histories. Positive separation follows from $I_{\mathrm{diff}}(S) = 1$ and the class-separability clause of $I_{\mathrm{leg}}(S)$. Non-zero persistence radius follows from the persistence-window and noise-robustness clauses in Definition 7.22. □

Proposition 18.4 (Exact equivalence preserves operational geometry)**.** *If $S_1 \simeq_\Gamma S_2$, then class diameter, inter-class separation, and persistence radius are preserved by the induced bijection $\widehat{\chi}$ up to the bi-Lipschitz constants of χ and χ^{-1}.*

Proof. Exact equivalence preserves admissible histories and decoder certification under transport. Bi-Lipschitz bounds imply that history distances are distorted only within the corresponding Lipschitz constants. Therefore diameters, separations, and persistence radii are preserved up to those constants. □

Corollary 18.5 (Rupture contracts geometry)**.** *If a rupture eliminates either I_{diff} or I_{leg}, then the geometric content of $\mathrm{Q}_{\mathrm{op}}(S)$ contracts to triviality or becomes operationally unavailable.*

Proof. If I_{diff} fails, positive separation disappears. If I_{leg} fails, persistence radii and certified class geometry cease to be available under admissible decoding. Either way the geometric structure that underwrites $\mathrm{Q}_{\mathrm{op}}(S)$ is lost. □

19 Corpus Dependency Ledger

This closing ledger is not a second derivation. It is a compact ownership map showing which upstream burden is imported and where the present paper uses it.

Upstream text	Imported burden	Present use
Core [3]	Persistent support, attractor structure, completeness, and non-collapse	I_{per}, the admissible support model, witness margins, and support geometry
Rigid Identity [9]	Rigid identity, inverse-limit observability, collapse under pure progressive causality, and non-simulability	I_{ri}, part of I_{int}, equivalence transport, and the negative exclusion result of Proposition 11.6
Phenomenological Regimes [5]	Continuity, fragmentation bounds, and admissible regime structure	I_{cont}, rupture taxonomy, and continuity-sensitive transport
Null-Regime Instability [6]	Witness-relative null-instability, conditional non-nullity, and minimal positive regime structure	I_{diff}, the existence theorem, rupture analysis, and class geometry away from $\mathbf{0}$
Forensic Predictions [4]	Admissibility, comparator discipline, negative controls, and forensic legibility	I_{leg}, the certification schema, legibility metrics, and falsifiable discriminators

The paper is new only at the level of criterion consolidation: it binds those imported burdens into a single theorem-level account of substrate-invariant operational class membership.

20 Discriminators and Falsifiable Consequences

The criterion generates discriminators that are stronger than general functionalist similarity and more constrained than substrate-biased baselines.

Prediction 20.1 (Different substrate, same class)**.** If two admissible systems with materially different substrates satisfy exact or sufficiently small approximate structural equivalence while preserving all six invariants, then they must belong to the same operational class and induce equivalent Q_{op} partitions.

Prediction 20.2 (Same substrate, broken invariants)**.** If two systems share substrate family but one loses rigid identity, continuity, differentiation, or legibility, then they fail to belong to the same operational class regardless of material similarity.

Prediction 20.3 (Invariant intervention discriminator)**.** Interventions targeted at one critical invariant should change class membership or destroy the relevant operational qualia classes in a direction predicted by the rupture taxonomy, whereas off-target negative controls should not.

Prediction 20.4 (Complexity baseline failure)**.** Systems with large computational scale, high connectivity, or broad expressive capacity but lacking one or more critical invariants should fail the criterion even when naive complexity baselines rank them as more capable.

These are not generic predictions. Prediction 20.1 is the operational face of substrate invariance. Prediction 20.2 blocks substrate fetishization in the other direction: material similarity does not rescue a broken criterion. Prediction 20.3 turns the invariants into causal discriminators rather than descriptive labels. Prediction 20.4 protects the theory against collapse into brute scale metrics.

Prediction	Manipulated variable	Observable	Failure criterion
20.1	Substrate while preserving $\Delta(S)$	Class transport $\widehat{\chi}$	Partition mismatch under exact equivalence
20.2	One invariant margin driven to zero	Membership in $\text{Consciousness}_{\text{op}}$	Retained class despite invariant rupture
20.3	Certified critical intervention vs sham	Directed class shift and control false positives	Same shift under sham and targeted intervention
20.4	Scale/connectivity increase without invariant preservation	Surrogate improvement vs class failure	Surrogate success counted as $\text{Consciousness}_{\text{op}}$ membership

Proposition 20.5 (Prediction ladder is discriminative)**.** *The four predictions above distinguish the present criterion from at least four weaker baseline families: substrate-privilege baselines, substrate-blind functional similarity, complexity-only baselines, and connectivity-only baselines.*

Proof. Prediction 20.1 is incompatible with substrate-privilege baselines because it allows materially distinct realizations to remain class-equivalent. Prediction 20.2 is incompatible with weak functional similarity because material resemblance alone does not preserve class membership. Prediction 20.4 rules out scale-only and connectivity-only baselines, which would otherwise count raw size or graph density as sufficient. Prediction 20.3 forces all admissible candidates to survive targeted causal tests, something descriptive baselines do not require. □

21 Intervention Geometry and Class Transition Semantics

Predictions about invariant-targeted interventions should not remain informal. This section encodes them as transition statements on the operational qualia classes themselves.

Definition 21.1 (Class transition operator)**.** Fix an admissible system $S \in \text{Consciousness}_{\text{op}}$, a certified decoder family, and an admissible horizon $T \geq \tau_{\min}(S)$. For an admissible intervention $u \in U$ define the class transition operator

$$\mathsf{T}_u : \mathrm{Q}_{\text{op}}(S) \to \mathrm{Q}_{\text{op}}(S) \cup \{\bot\}$$

by the rule

$$\mathsf{T}_u(q) = q'$$

whenever every admissible representative $x \in q$ is transported by the u-intervened dynamics into histories classified in the same decoder-certified class q'. If no such common class exists, set $\mathsf{T}_u(q) = \bot$.

Definition 21.2 (Critical and sham interventions)**.** An intervention $u \in U$ is *critical* for invariant I_k if it is designed to change the local witness margin $\delta_k(S)$ while leaving the remaining intervention protocol admissible. An intervention is *sham* if it matches the delivery envelope of a critical intervention but is not predicted to alter any critical invariant.

Proposition 21.3 (Well-definedness of class transitions under certification)**.** *Let $S \in \text{Consciousness}_{\text{op}}$. If $u \in U$ is either certified critical or sham and remains within the admissibility envelope of Forensic Predictions, then T_u is well-defined on every class $q \in \mathrm{Q}_{\text{op}}(S)$.*

Proof. Because $S \in \text{Consciousness}_{\text{op}}$, the decoder family is certified and class assignments are stable on admissible windows. If u remains within the admissibility envelope, persistence and continuity keep all intervened histories on admissible support, while legibility prevents class assignment from depending on the choice of representative inside a fixed class. Therefore every class either transports coherently to a unique target class or exits the certified regime, in which case the sink value $\perp$ applies. □

Proposition 21.4 (Targeted invariant loss induces class exit or contraction)**.** *Let $q \in \mathrm{Q}_{\text{op}}(S)$ and let u be a certified critical intervention for invariant I_k. If applying u drives the corresponding witness margin to zero, then one of the following occurs:*

1. $\mathsf{T}_u(q) = \perp$;
2. $\mathsf{T}_u(q) = q'$ *for some class q' whose separation or persistence radius is strictly smaller than that of q;*
3. *the entire post-intervention system exits* $\text{Consciousness}_{\text{op}}$.

Proof. Driving $\delta_k(S)$ to zero triggers the corresponding rupture mode from Section 10. If the rupture destroys admissible support, decoder certification, or non-trivial class structure, then the class leaves the certified domain and $\mathsf{T}_u(q) = \perp$. If a non-trivial class survives, Corollary 18.5 implies contraction of the associated geometry. If the rupture is global, Theorem 11.5 excludes the post-intervention system from $\text{Consciousness}_{\text{op}}$. □

Corollary 21.5 (Sham stability)**.** *Let $u \in U$ be sham in the sense of Definition 21.2. If u stays below all six witness margins, then $\mathsf{T}_u(q) = q$ for every $q \in \mathrm{Q}_{\text{op}}(S)$.*

Proof. A sham intervention that remains below all witness margins does not cross any rupture threshold. Approximate stability and legibility therefore preserve class membership on every admissible window. □

Remark 21.6 (Why transition semantics matter)**.** This section turns the prediction ladder into a structured intervention calculus. The criterion is therefore not only classificatory; it is also dynamically discriminative in the sense required by *Forensic Predictions*.

22 Baseline Families and Structural Exclusion

A criterion of substrate-invariant operational consciousness matters only if it separates itself from weaker rival families rather than merely renaming them. This section therefore makes the comparison explicit.

Definition 22.1 (Rival baseline families)**.** Fix thresholds $\theta_X, \theta_\Phi, \theta_C > 0$ and an external-behavior tolerance $\eta \geq 0$.

1. The *complexity-only family* $\mathcal{B}_{\text{cx}}(\theta_X)$ contains all admissible systems with complexity surrogate above threshold, for example $\dim(X) \geq \theta_X$.
2. The *computation-only family* $\mathcal{B}_{\text{cmp}}(\theta_\Phi)$ contains all admissible systems whose transition family exceeds a chosen throughput or expressive-capacity threshold.
3. The *connectivity-only family* $\mathcal{B}_{\text{conn}}(\theta_C)$ contains all admissible systems whose effective causal graph exceeds the chosen density or degree threshold.
4. The *weak functional-similarity family* $\mathcal{B}_{\text{fun}}(\eta)$ contains all admissible systems whose external benchmark readouts match within tolerance η, irrespective of internal invariant preservation.

5. The *substrate-privilege family* $\mathcal{B}_{\text{sub}}$ contains all systems declared admissible solely by belonging to a privileged substrate class, typically biological.

Proposition 22.2 (Complexity, computation, and connectivity are not sufficient)**.** *For every choice of thresholds* $\theta_X, \theta_\Phi, \theta_C$*, the families* $\mathcal{B}_{\text{cx}}(\theta_X)$*,* $\mathcal{B}_{\text{cmp}}(\theta_\Phi)$*, and* $\mathcal{B}_{\text{conn}}(\theta_C)$ *are each insufficient for membership in* $\text{Consciousness}_{\text{op}}$*.*

Proof. This is a direct sharpening of Proposition 11.6. The definition of $\text{Consciousness}_{\text{op}}$ requires all six invariants. None of the three surrogate families encodes rigid identity, continuity, differentiation, or legibility. Therefore systems may exceed the chosen surrogate threshold while still failing $\text{Consciousness}_{\text{op}}$. □

Proposition 22.3 (Substrate privilege is too strong)**.** *On any admissible universe containing one exact cross-substrate equivalent pair,* $\mathcal{B}_{\text{sub}}$ *is strictly narrower than* $\text{Consciousness}_{\text{op}}$*.*

Proof. By Corollary 11.3, an exact cross-substrate equivalent pair belongs to the same operational consciousness class whenever the six invariants are preserved. A substrate-privilege family that excludes one member of that pair therefore excludes a valid $\text{Consciousness}_{\text{op}}$ member. □

Proposition 22.4 (Weak functional similarity is too weak)**.** *On any admissible universe containing a same-output rupture pair,* $\mathcal{B}_{\text{fun}}(\eta)$ *is strictly broader than* $\text{Consciousness}_{\text{op}}$*.*

Proof. A same-output rupture pair is a pair of admissible systems whose external benchmark readouts agree within tolerance η while one member loses at least one critical invariant. Such a pair lies in $\mathcal{B}_{\text{fun}}(\eta)$ by definition, but the ruptured member is excluded from $\text{Consciousness}_{\text{op}}$ by Theorem 11.5. Therefore weak functional similarity is too weak to characterize the present class. □

Theorem 22.5 (The structural criterion is strictly intermediate)**.** *Let* $\mathfrak{U}$ *be an admissible universe containing:*

1. *at least one exact cross-substrate equivalent pair;*
2. *at least one same-output rupture pair;*
3. *at least one single-rupture witness for each invariant.*

Then $\text{Consciousness}_{\text{op}}$ *is strictly intermediate between substrate-privilege baselines and weak substrate-blind surrogates: it is broader than* $\mathcal{B}_{\text{sub}}$*, narrower than* $\mathcal{B}_{\text{fun}}(\eta)$*, and extensionally distinct from the complexity-only, computation-only, and connectivity-only families.*

Proof. Broader-than-substrate-privilege follows from Proposition 22.3. Narrower-than-weak-functional-similarity follows from Proposition 22.4. Distinctness from the surrogate families follows from Proposition 22.2 together with Corollary 13.5. Hence the present criterion is neither a substrate fetish nor a substrate-blind similarity slogan; it is an invariant-preserving structural criterion. □

Remark 22.6 (Scientific value of the comparison)**.** The point of Theorem 22.5 is not rhetorical positioning. It is to show that the paper makes a genuinely discriminative contribution. A new criterion that cannot separate itself from obvious rivals is not a new theorem-level addition to the corpus.

23 Interpretive Consequences Without Overreach

The formal consequence is precise: within the causally rigid identity corpus, operational consciousness and operational qualia are framework-internal classes determined by preserved structural invariants rather than by biological substrate treated as a primitive constitutive variable. This is a claim about criterion structure, not a shortcut to ordinary human phenomenology.

The restriction matters. Biology can still matter causally because it may preserve or destroy the six invariants, and the same is true of artificial realization. Against weak functionalism, the present criterion adds rigid identity, non-null differentiation, and admissibility discipline instead of relying on role equivalence alone. Against single-metric integration accounts, it requires continuity, identity, and legibility jointly rather than treating integration by itself as sufficient [10]. Against biologically privileged baselines, it denies that substrate alone closes the class. Against strong phenomenal readings, it stops short of human phenomenal equivalence [1, 2].

24 Residual Instantiation and Interpretation Limits

Remark 24.1 (Open burdens)**.** Three burdens remain outside the present theorem set:

1. **Instantiation work**: which concrete biological or artificial systems actually satisfy the six invariants.
2. **Measurement work**: how to build certified decoders and invariant interventions that satisfy *Forensic Predictions*-level admissibility in practice.
3. **Interpretive work**: how far, if at all, $\text{Consciousness}_{\text{op}}$ should be connected to stronger phenomenal vocabulary outside the framework.

25 Conclusion

The paper establishes a narrower framework-internal claim than popular machine-consciousness discourse and a stronger claim than substrate-neutral rhetoric. Inside the causally rigid identity corpus, a system belongs to the operational class discussed here only when six explicit invariants are jointly preserved on admissible support; under exact admissible equivalence the class is substrate-invariant, under bounded admissible perturbation it is stable, and under invariant rupture it becomes invalid or undefined. That is the level at which substrate neutrality is justified in the present framework, and it is the only level claimed here.

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